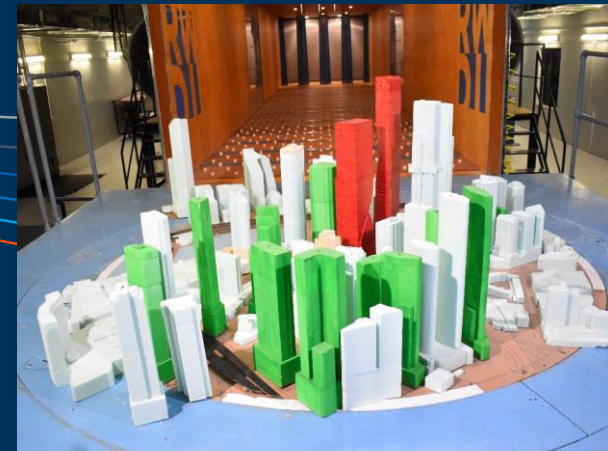


Wind tunnel testing and wind engineering

University of Queensland

Graeme Wood
29 April 2026



What does wind engineering impact?

External wind speed and turbulence – Airport Planning, Architecture, Asset Management, Bridge Engineering, Building Physics, Civil Engineering, Façade, Fire Engineering, Geotechnical, Landscape Architecture, Maritime, Mechanical, Rail Engineering, Structural Design, Transportation, Urban design

Wind loading and response – AT&R, Bridge Engineering, Electrical, Facades, Lighting design, Structures, Transportation

Internal pressure – Architecture, Facades, Fire, Mechanical, Structures, Transportation, Tunnel design, Vertical transportation

Wind-driven rain – Architecture, Facades, Hydraulics

Wind-induced noise – Acoustic, Structural vibration

Wind climate – Facades, Planning, Risk, Structures

Pollution dispersal – Mechanical, Public Health

A series of overlapping, wavy lines in shades of blue and orange, flowing from left to right across the bottom of the slide.

Wind-tunnel vs CFD



physical

vs

numerical

- Models turbulence well at limited speed
 - Good benchmarking
 - Long-time to build model, but quick to alter and retest
 - Difficult to model all turbulence scales in the wind-tunnel
- Good for internal flow or MEAN flows – not for PEAK response (unless using computationally heavy solvers)
 - Architectural models generally needs rebuilding
 - Any geometry changes requires complete re-run of the model
 - Need to benchmark results

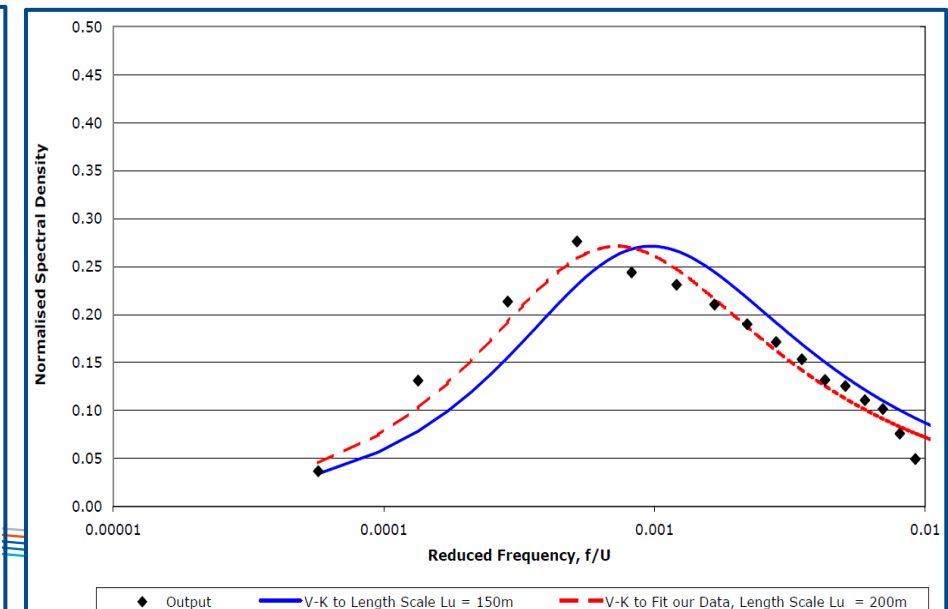
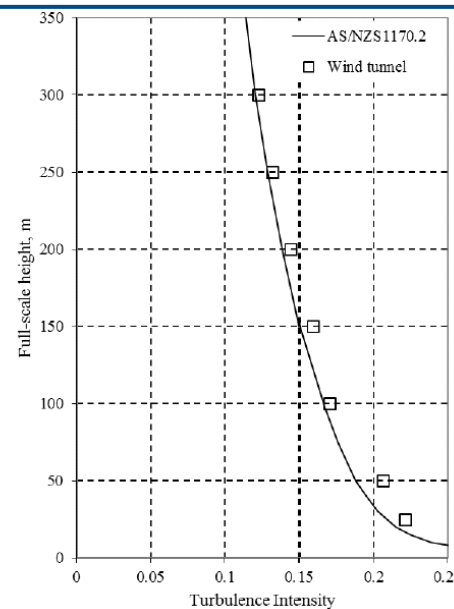
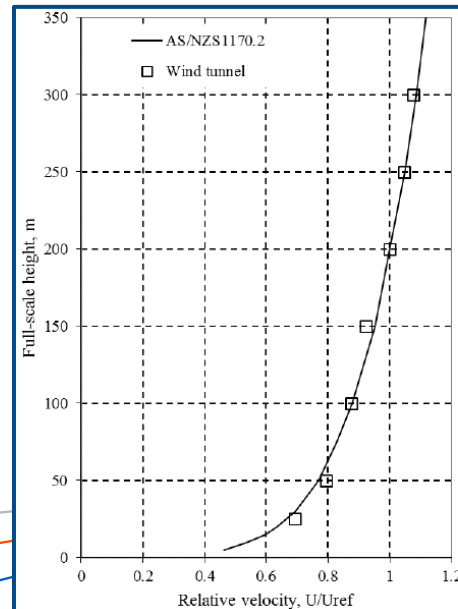
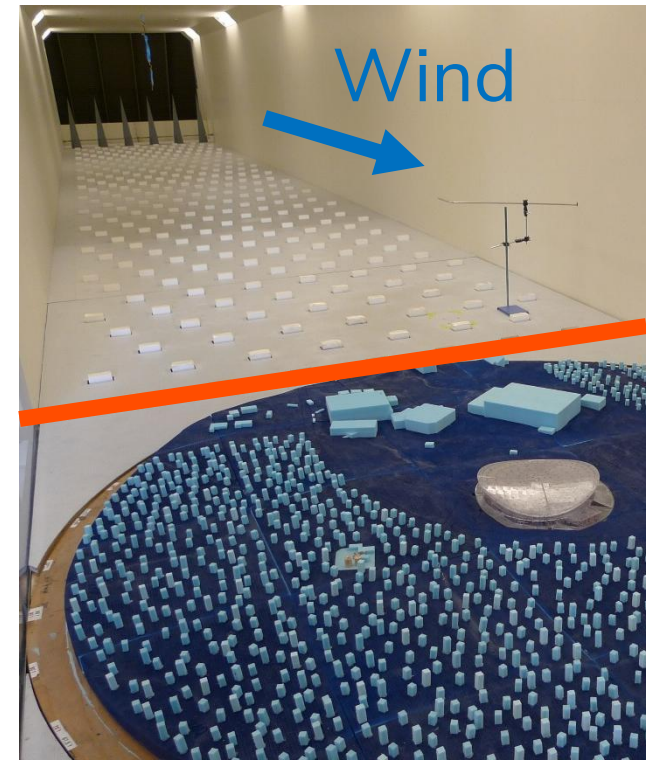
Wind-tunnel scaling

Fundamental height distribution of:

- Mean velocity
 - Turbulence intensity
 - Spectrum of wind energy within a factor of 3
- } within 10%

Optional

- Structural dynamic properties



Scaling

Reynolds number scaling

$$RN_{\text{model}} = RN_{\text{proto}} = \frac{\rho_m v_m L_m}{\mu_m} = \frac{\rho_p v_p L_p}{\mu_p}$$

ρ air density, kg/m³

v velocity, m/s

L length, m

μ dynamic viscosity, Ns/m²

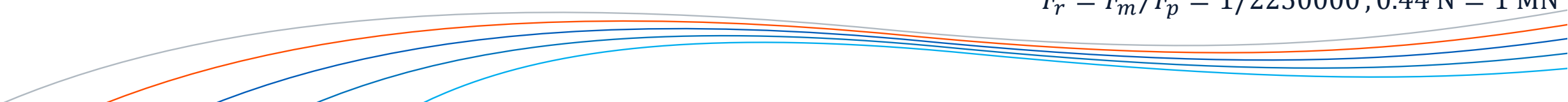
in wind tunnel $\rho_m = \rho_p$, and $\mu_m = \mu_p$

velocity and length scale ratios

$$V_{r(\text{atio})} = v_m/v_p \quad L_r = L_m/L_p \quad (\text{or } v_r = 1/L_r)$$

gives time, frequency, mass, force, moment...ratios

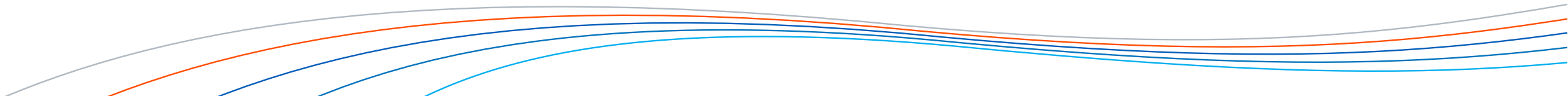
$$T_r = L_r/v_r \quad f_r = 1/T_r \quad m_r = L_r^3 \quad F_r = L_r^2 \cdot v_r^2 \quad M_r = L_r^3 \cdot v_r^2 \quad \text{Example}$$

$$\text{Froude number, } Fr = \frac{v}{\sqrt{gL}} \quad \text{Mach number, } Ma = \frac{v}{c} \quad \dots$$
$$L_r = L_m/L_p = \frac{1}{500}, 1 \text{ mm} = 0.5 \text{ m}$$
$$v_r = v_m/v_p = 1/3, 10 \text{ m/s} = 30 \text{ m/s}$$
$$t_r = t_m/t_p = 1/167, 3.6 \text{ s} = 10 \text{ min}$$
$$F_r = F_m/F_p = 1/2250000, 0.44 \text{ N} = 1 \text{ MN}$$


Wind tunnel tests

Combine results with wind climate analysis

- flow visualisation
- wind speed studies (ped wind/aircraft)
- pressure studies
- structural tests:
 - stick aeroelastic models
 - high frequency base balance studies
 - simultaneous pressure studies
 - aeroelastic studies (buildings, bridges)
- air quality
- wind induced noise
- benchmark for CFD analysis



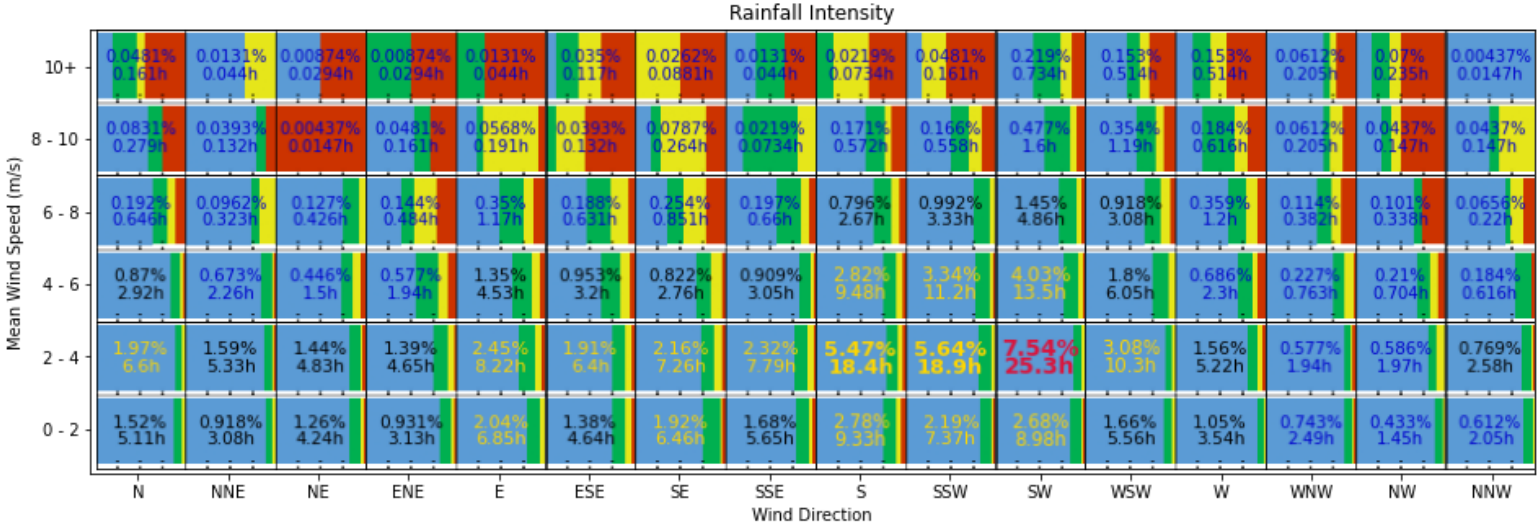
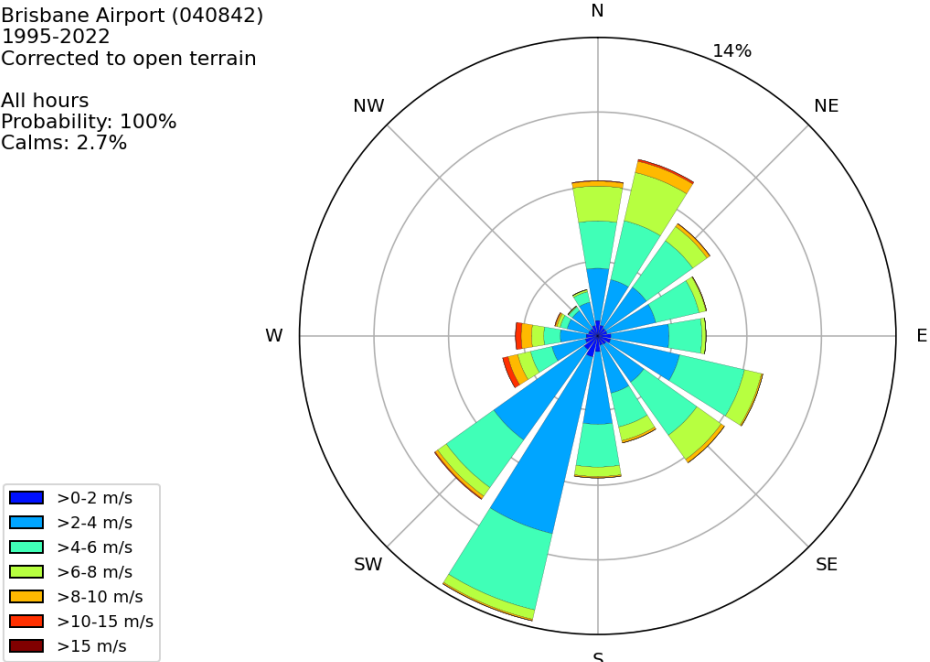
Climate analysis



- Check and clean data
- Correct wind data for roughness, terrain, height

Brisbane Airport (040842)
1995-2022
Corrected to open terrain

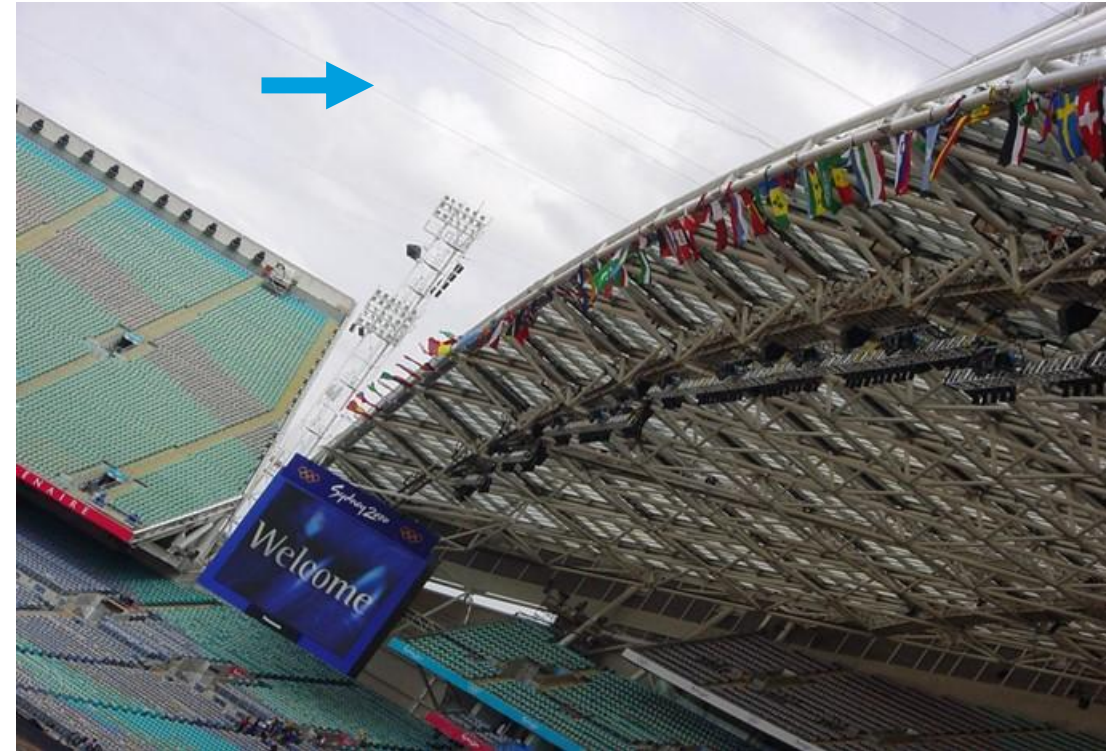
All hours
Probability: 100%
Calms: 2.7%



Data from Badgerys Creek, 1996 - 2022, BoM station No. 067108
10 minute data pre-2017, and 30 minute data post-2017 is
Rain occurs for 3.83% of the time, 335 hours/annum
Calm for 6.73% of rain events, 22.6 hours/annum

Light, 0 - 0.05 mm/min, 78.6% of data, 264 hours/annum
Moderate, 0.05 - 0.1 mm/min, 13.4% of data, 44.9 hours/annum
Heavy, 0.1 - 0.2 mm/min, 5.05% of data, 16.9 hours/annum
Severe, 0.2 - 10 mm/min, 2.97% of data, 9.96 hours/annum

Flow visualisation

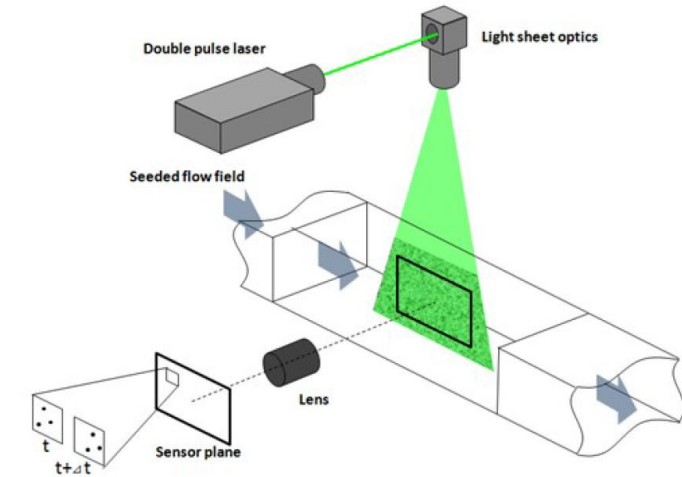
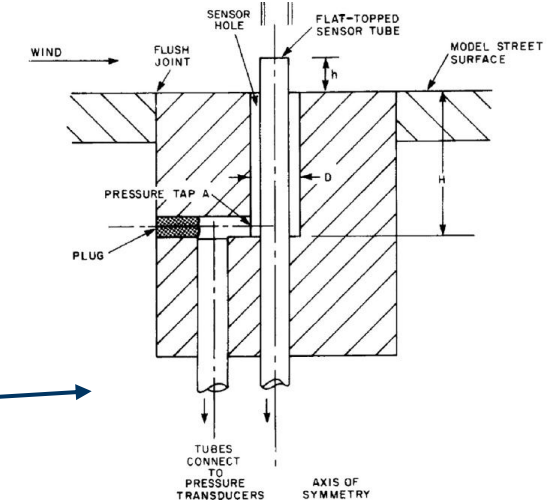
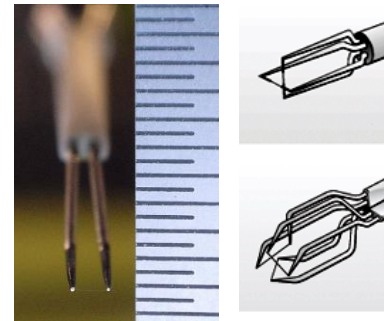


Can generally use CFD for flow visualisation

Measuring wind speed

Discrete point measurements

- Hot-wire / hot-film anemometry
 - Measures in plane orthogonal to wire
 - Susceptible to wind angle
 - Can only measure at few locations simultaneously
- Irwin probes
 - Pressure based
 - Poor in low-speed and high turbulence
 - Can measure at multiple locations simultaneously
- Multi-hole pressure sensor
 - Gives more information on directionality



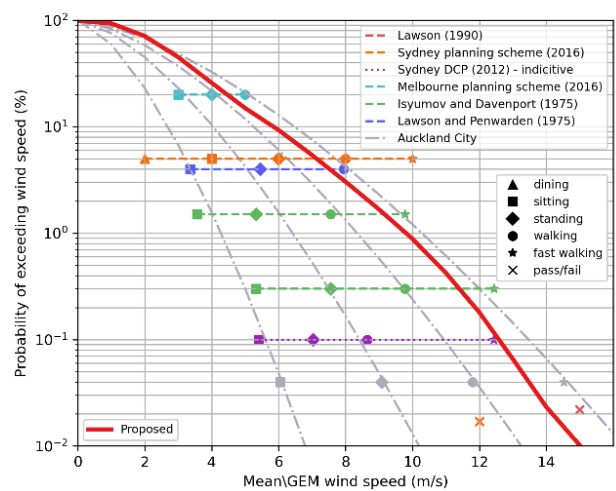
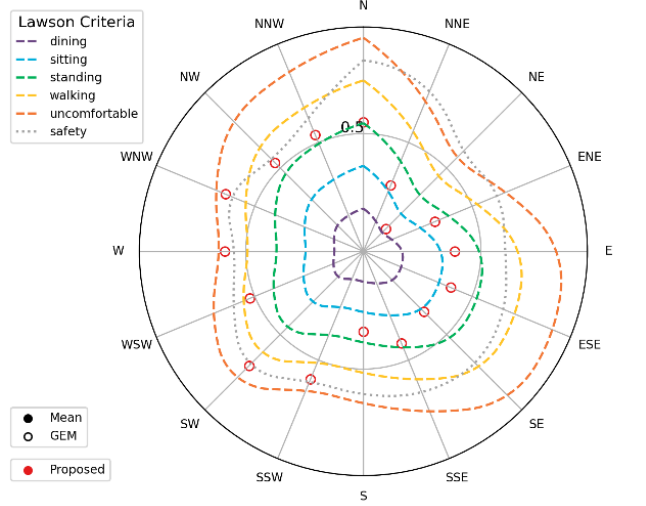
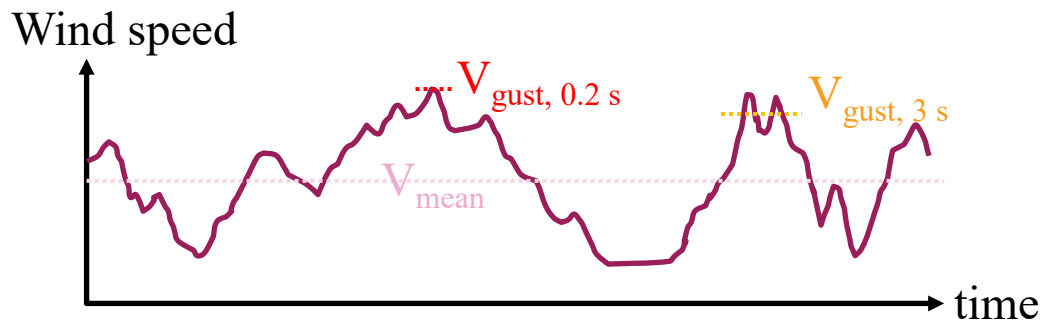
Flow-field measurements

- PIV
 - Difficult to measure in city-scapes
- Particle scour
 - Time-consuming and messy

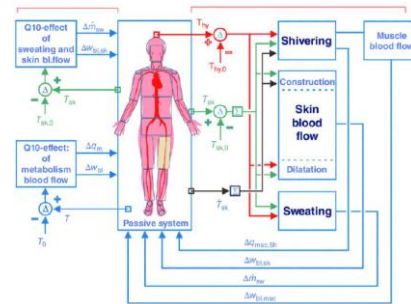


Can generally use CFD for **mean** wind speed

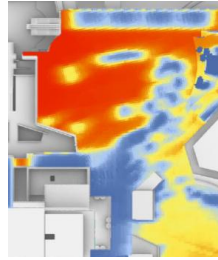
Pedestrian comfort



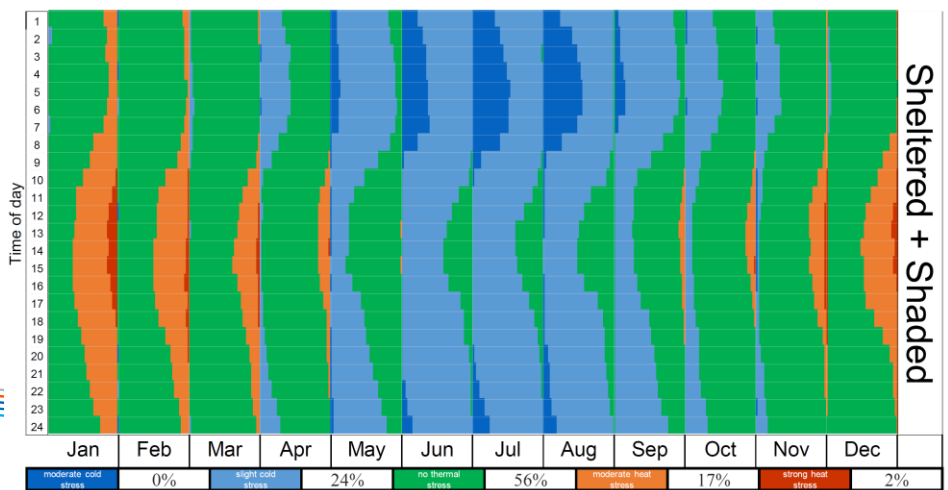
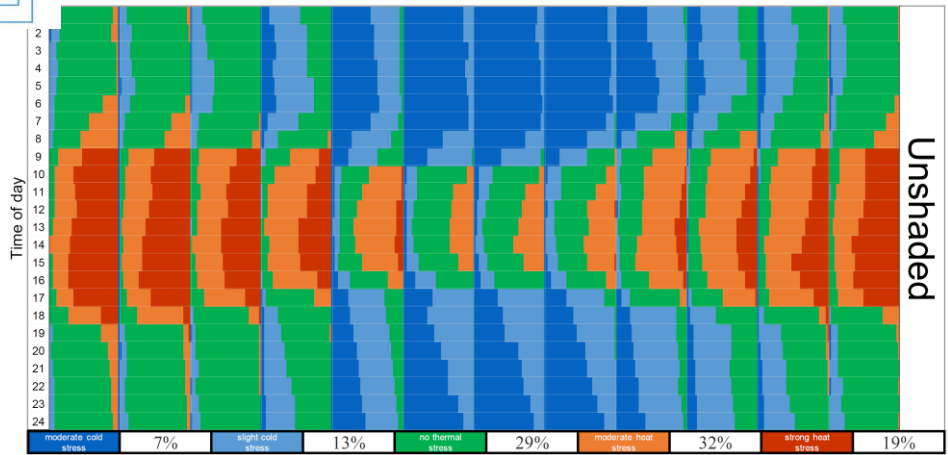
Fiala thermal comfort model



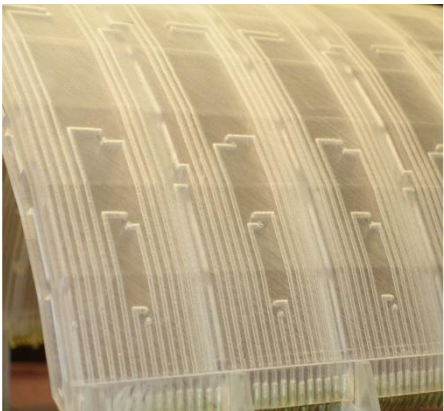
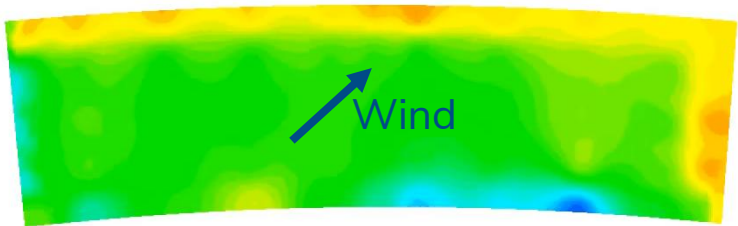
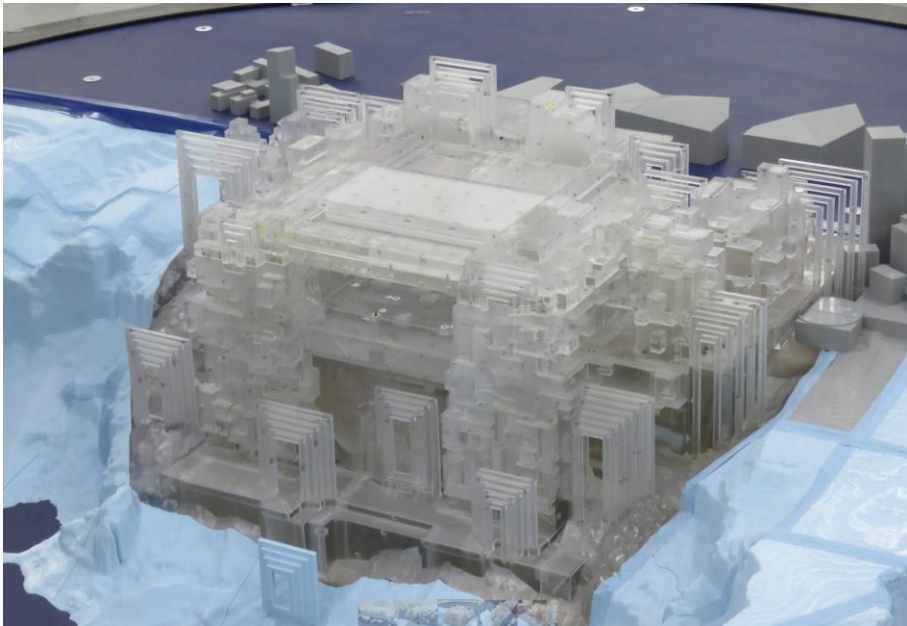
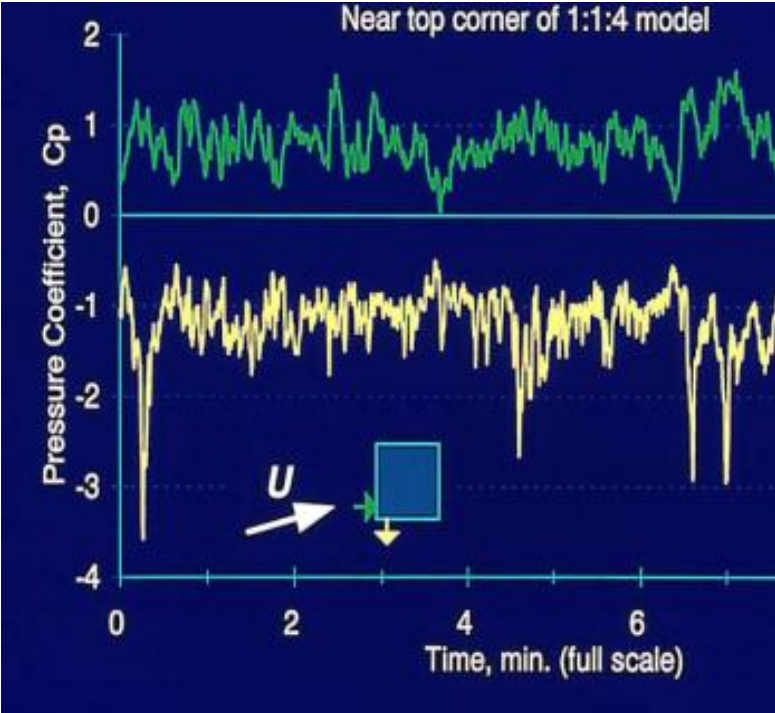
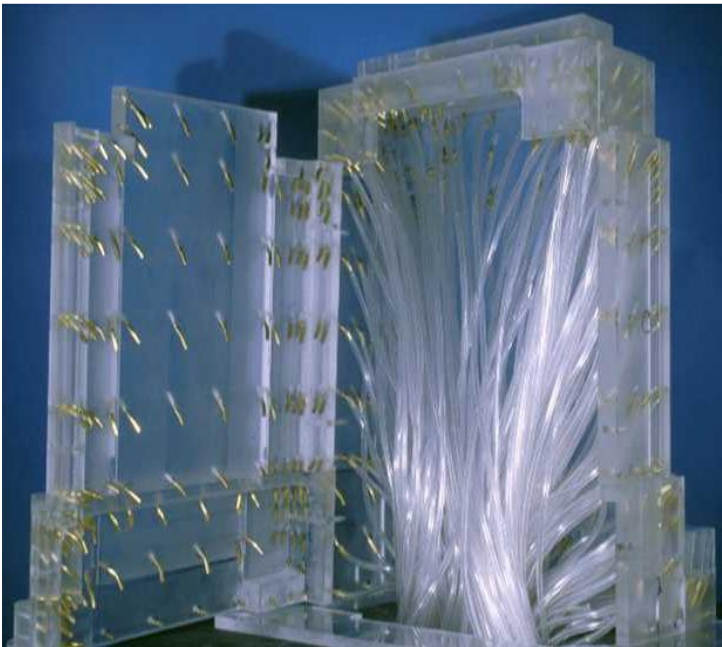
Solar exposure



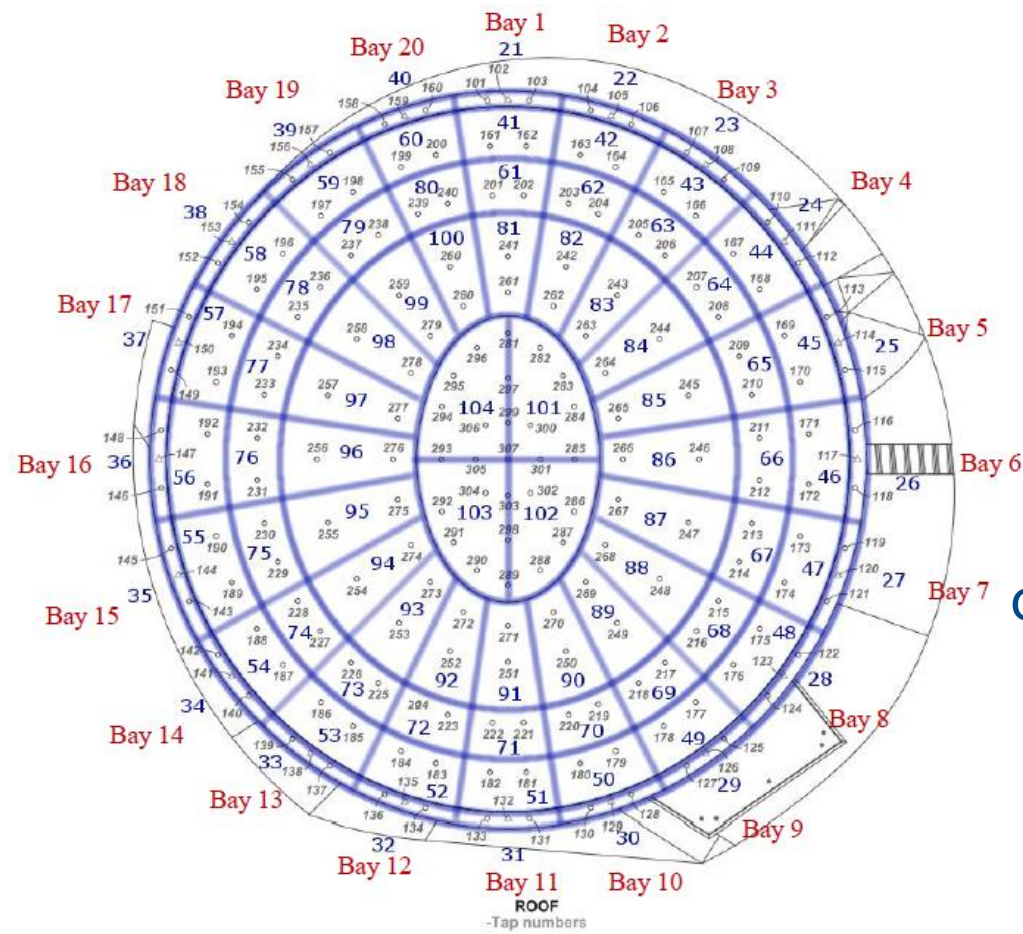
Wind speed



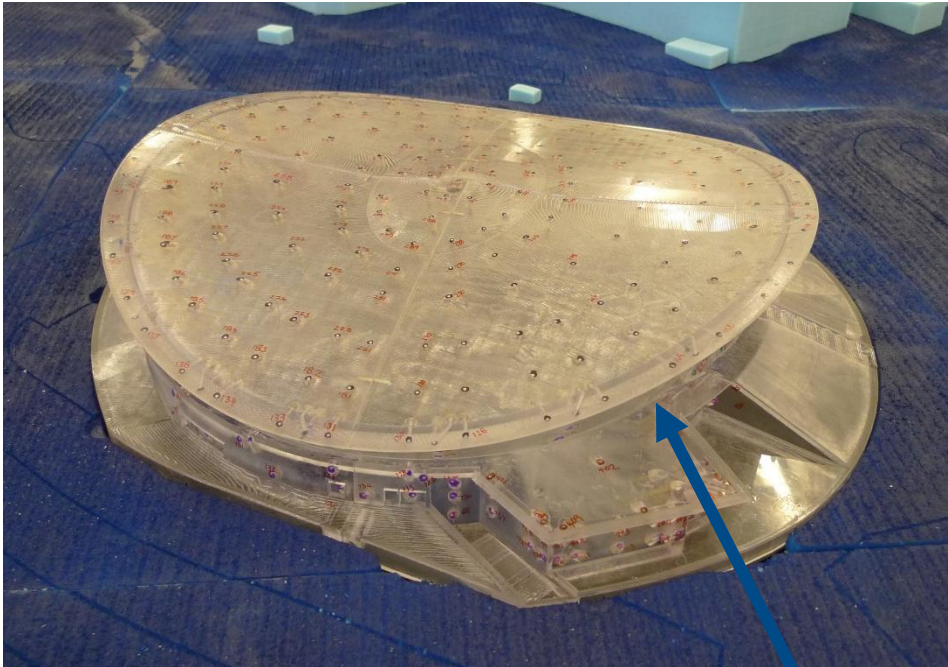
Pressure models



Queensland State Velodrome



Influence
coefficient
analysis

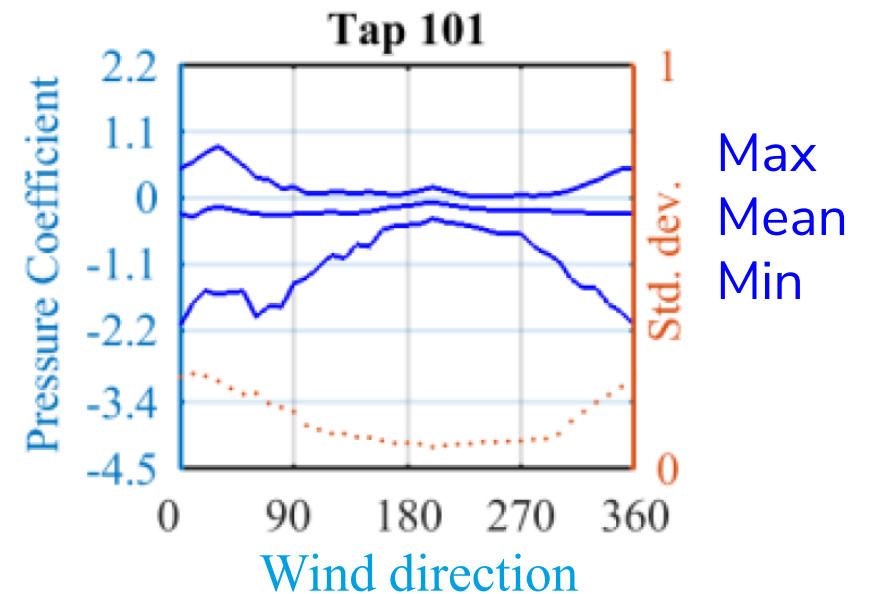
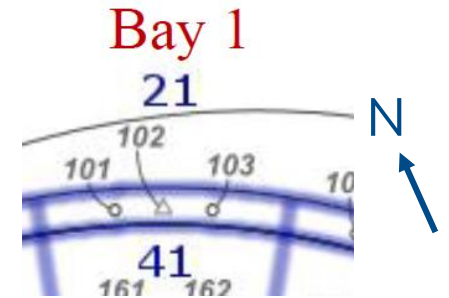


Operable vents
under eaves

Queensland State Velodrome

- Pressure tap layout designed with structural layout
- Simultaneous external pressure measurements for:
 - cladding design
 - quasi-steady structural design
 - mechanical systems
- Incorporate structural response

$$C_p = \frac{p}{0.5 \cdot \rho \cdot V_{ref}^2}$$



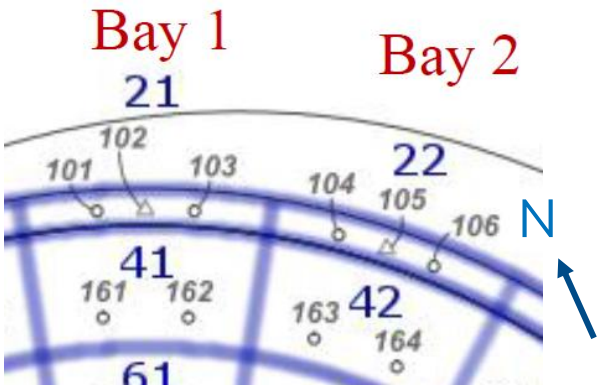
Queensland State Velodrome – cladding loads

Cladding

Tap	Max Coefficient		Max External Pressure		Max Net Pressure		Min Coefficient		Min External Pressure		Min Net Pressure	
	Cp	Dir (°)	Pe (kPa)	Dir (°)	Pt (kPa)	Dir (°)	Cp	Dir (°)	Pe (kPa)	Dir (°)	Pt (kPa)	Dir (°)
101	0.86	30	1.38	30	2.01	30	-2.12	0	-3.29	60	-3.29	60
102	1.56	20	2.41	20	3.03	30	-1.04	300	-1.43	80	-1.43	80
103	0.86	30	1.38	30	2.01	30	-2.32	0	-3.60	70	-3.60	70

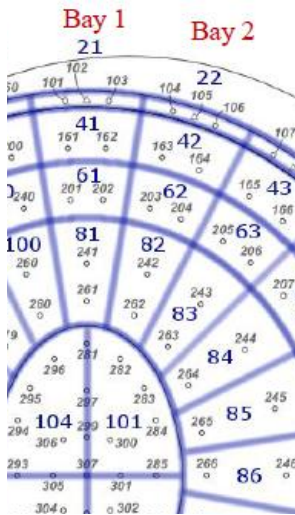
Cladding - differential

Virtual Tap	Tap A	Tap B	Max Coefficient		Max Net Pressure		Min Coefficient		Min Net Pressure	
			Cp	Dir (°)	Pt (kPa)	Dir (°)	Cp	Dir (°)	Pt (kPa)	Dir (°)
9001	101	102	0.93	310	1.26	30	-3.23	0	-4.35	0
9002	103	102	0.87	310	1.13	30	-3.36	0	-4.53	0
9003	104	105	1.58	350	2.32	30	-3.45	70	-5.62	70



Queensland State Velodrome – structural loads

Structural input



Influence coefficients

Bay	Panel	UPLIFT
		1
1	1	0.176
	21	0.161
	41	0.527
	61	0.377
	81	0.582
2	2	0.216
	22	0.153
	42	0.485
	62	0.363
	82	0.558

...

Response/kPa

•
•
•
104+internal

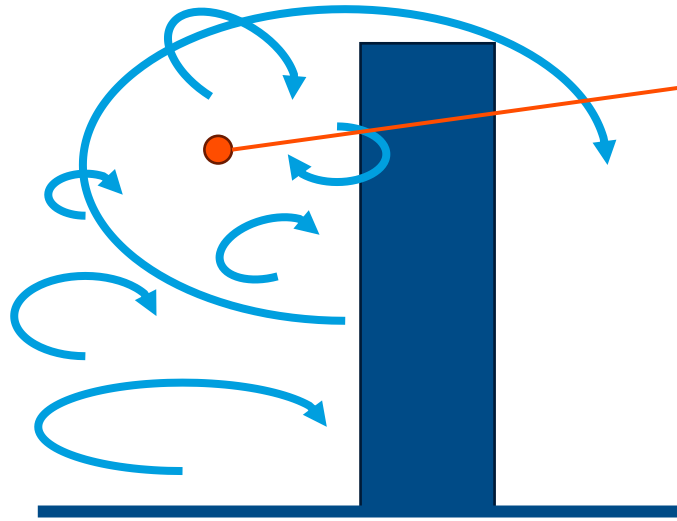
Structural output – 2 x influence coefficients

Load Case	Response	Direction (°)	Bay 1				
			Pressure acting on each panel (kPa)				
			1	21	41	61	81
1	5.368	320	-0.94	0.55	-0.27	-0.03	0.20
2	5.318	320	-0.94	0.55	-0.27	-0.03	0.20
3	3.150	10	0.62	-0.40	0.17	0.04	-0.13

••• 104+internal

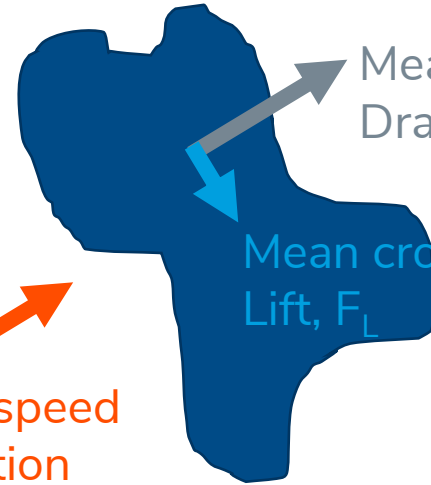
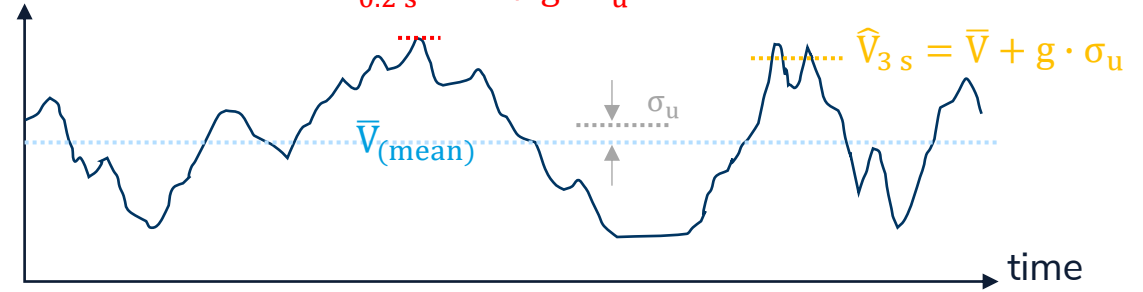
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Wind loading on tall buildings



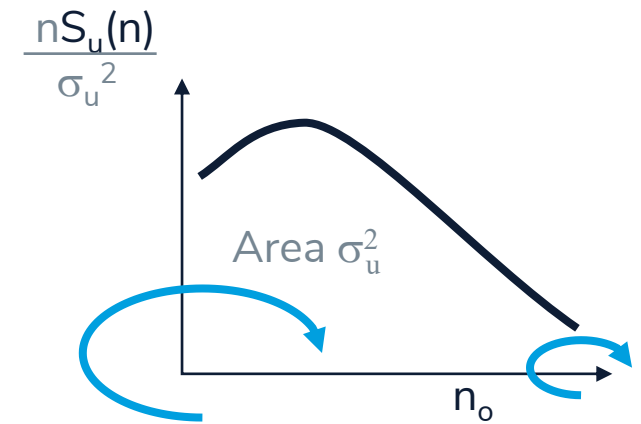
Mean wind speed
and direction

Wind speed



Mean along-wind
Drag, F_D

Mean cross-wind
Lift, F_L



Along wind excitation mechanisms

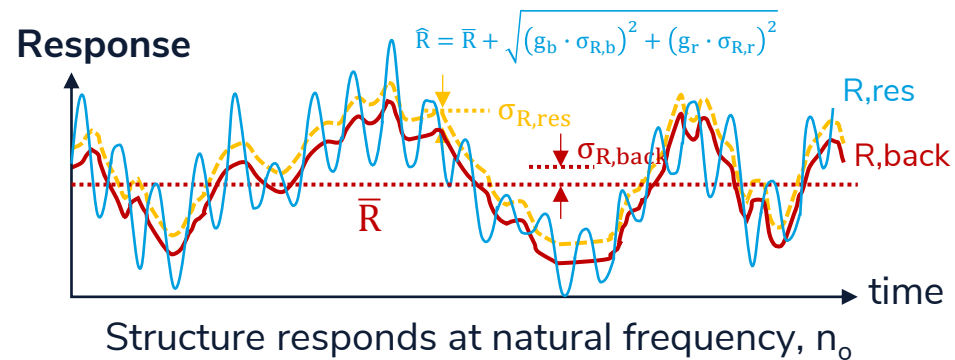
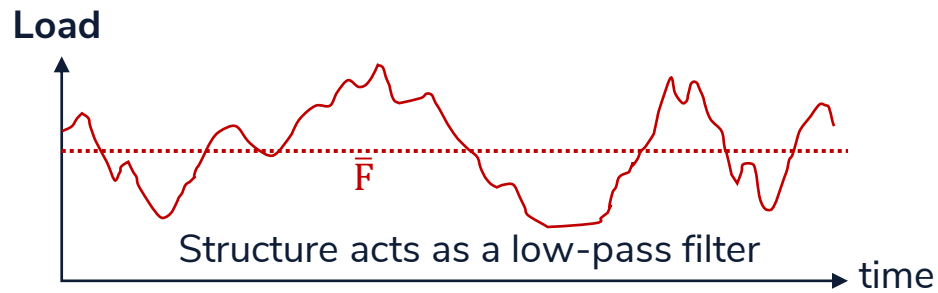
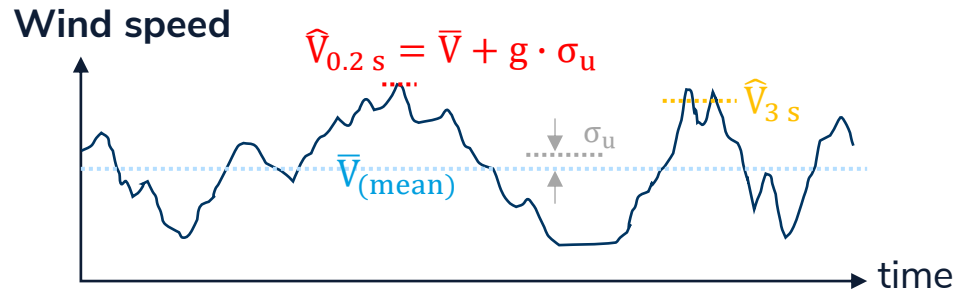
- Turbulence buffeting (background)

Cross wind excitation mechanisms

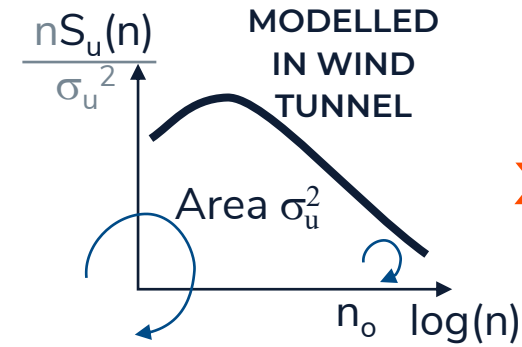
- Turbulence buffeting (background)
- Wake excitation (background/resonant)
- Motion dependent (background/resonant)

Fundamental methodology

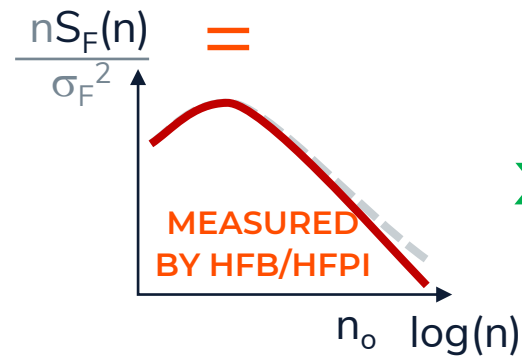
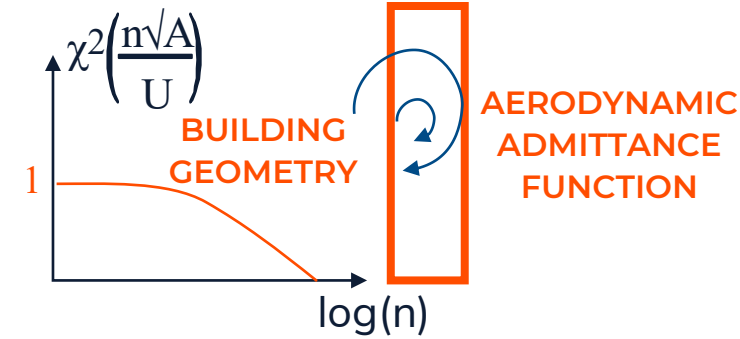
Time domain



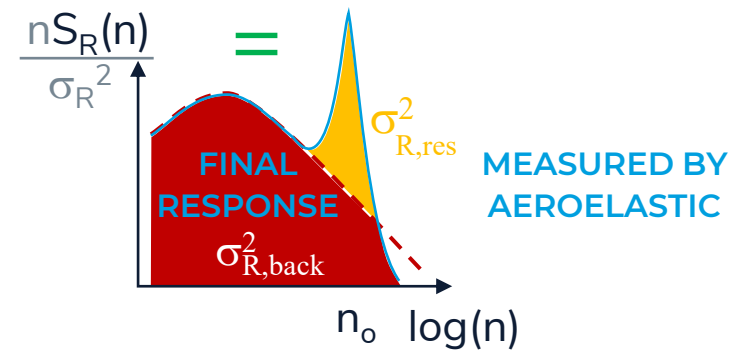
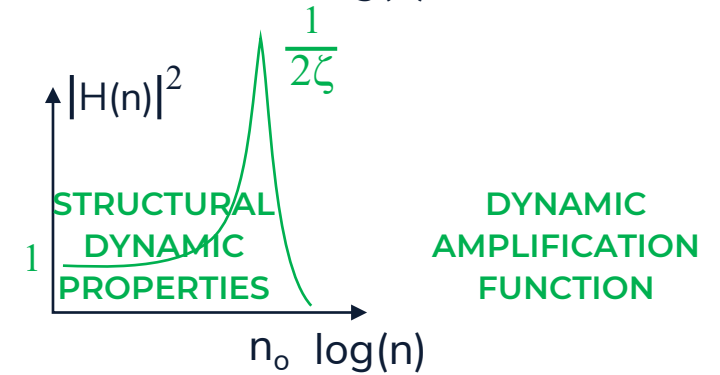
Frequency domain



X



X



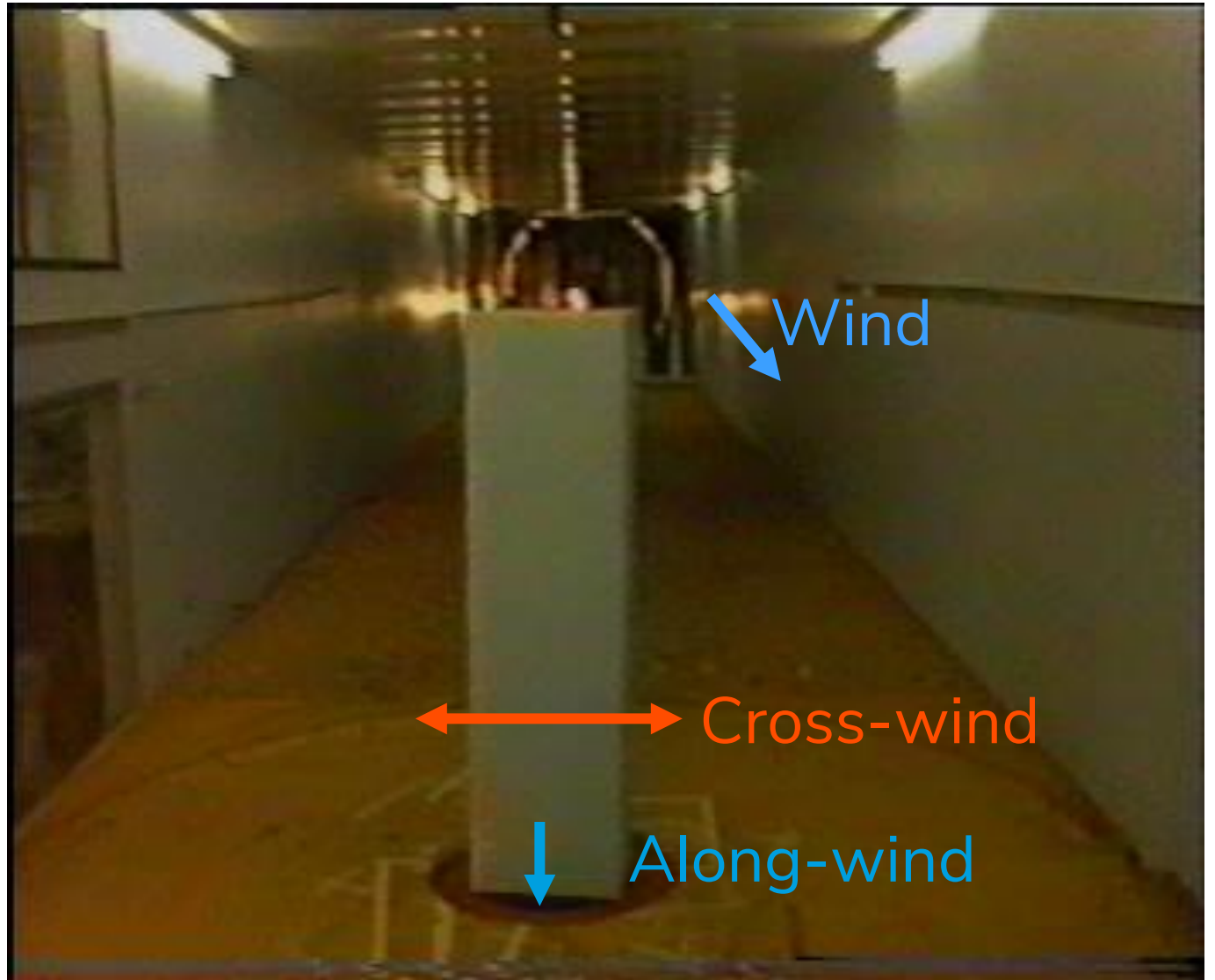
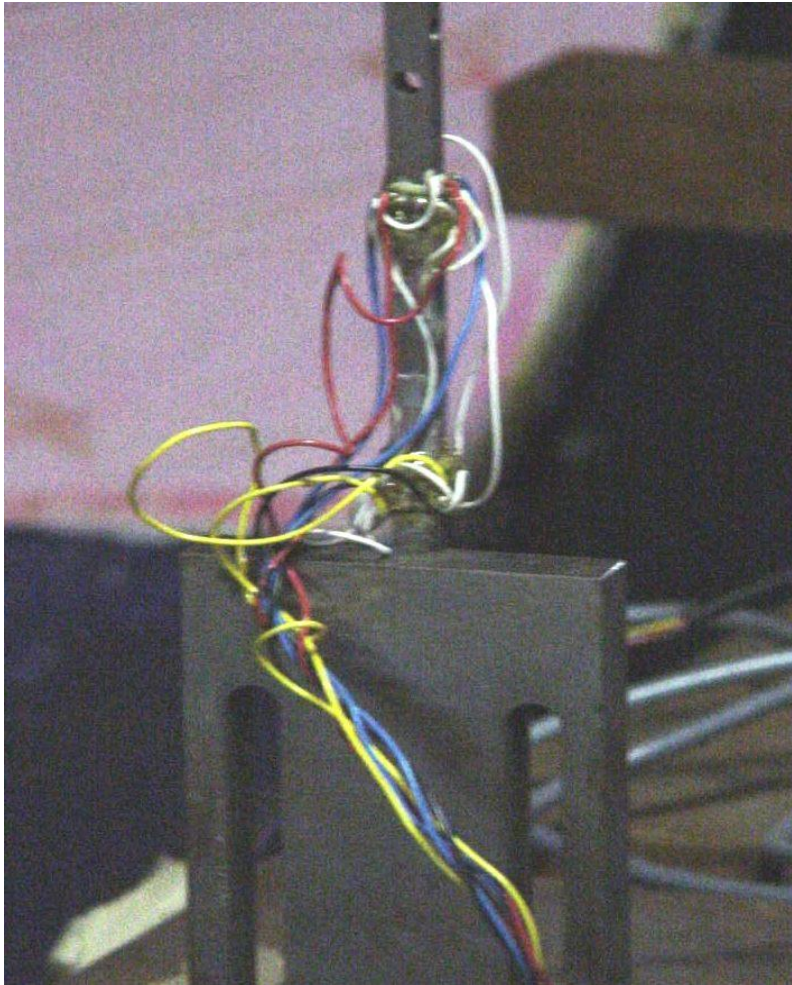
Tests for overall structural response of tall buildings



- Aeroelastic stick model (Response)
- High frequency balance (Force)
- Integrated surface pressures (Force)
- Multi-mode aeroelastic model (Response)
 - lumped mass
 - spine models
 - true aeroelastic models

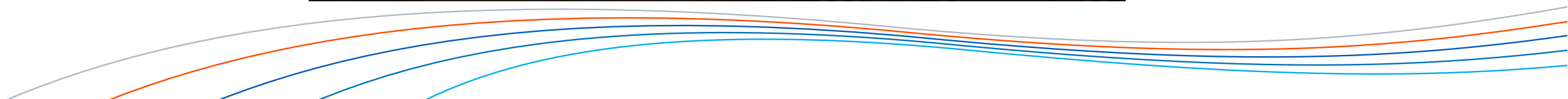
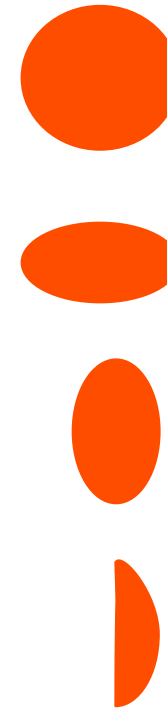


Aeroelastic stick model (Response)



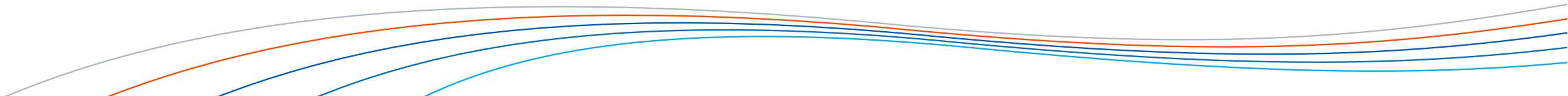
Cross-wind response: wake excitation

- Most common tall building excitation mechanism



Aeroelastic stick model

- Simple re-useable test rig
- Easy to incorporate damping
- Simplified mass and stiffness distribution
- Restricted to first translational mode in each direction, no torsion
- Long wind tunnel testing time
- Direct measure of mean and dynamic response, so no analysis, but can't change structural properties
- Mode shape corrections are required



High Frequency Balance



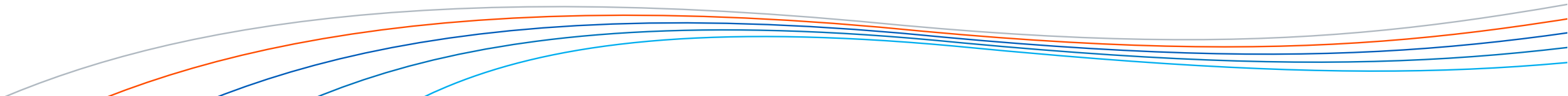
123 Albert St, Brisbane



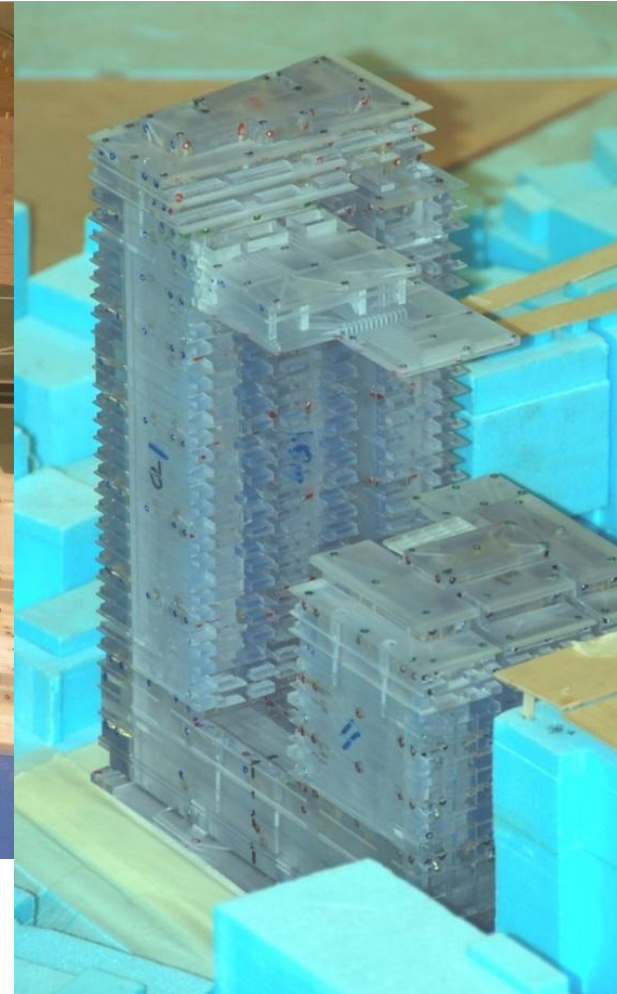
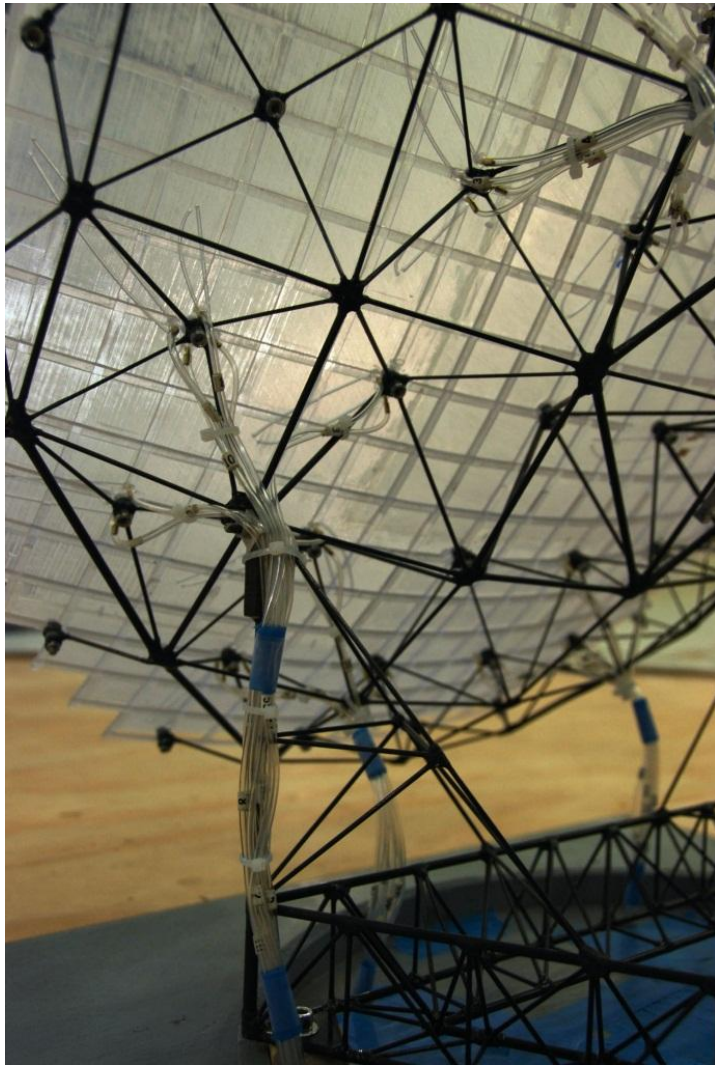
118 Mount Street,
Sydney

High Frequency Balance (Force)

- Rigid model, no attempt to model aeroelastic properties of the structure
- High natural frequency of model balance system (check in report should be >80 - 100 Hz)
- Generally only measures base responses
- Short testing time, but complex post processing for complex mode shapes
- Mode shape corrections are required
- Easy to recalculate response if structure changes

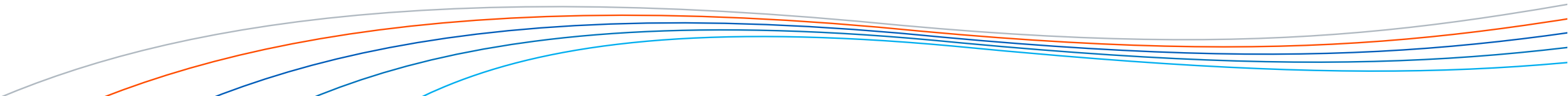


Integrated pressure (Force)

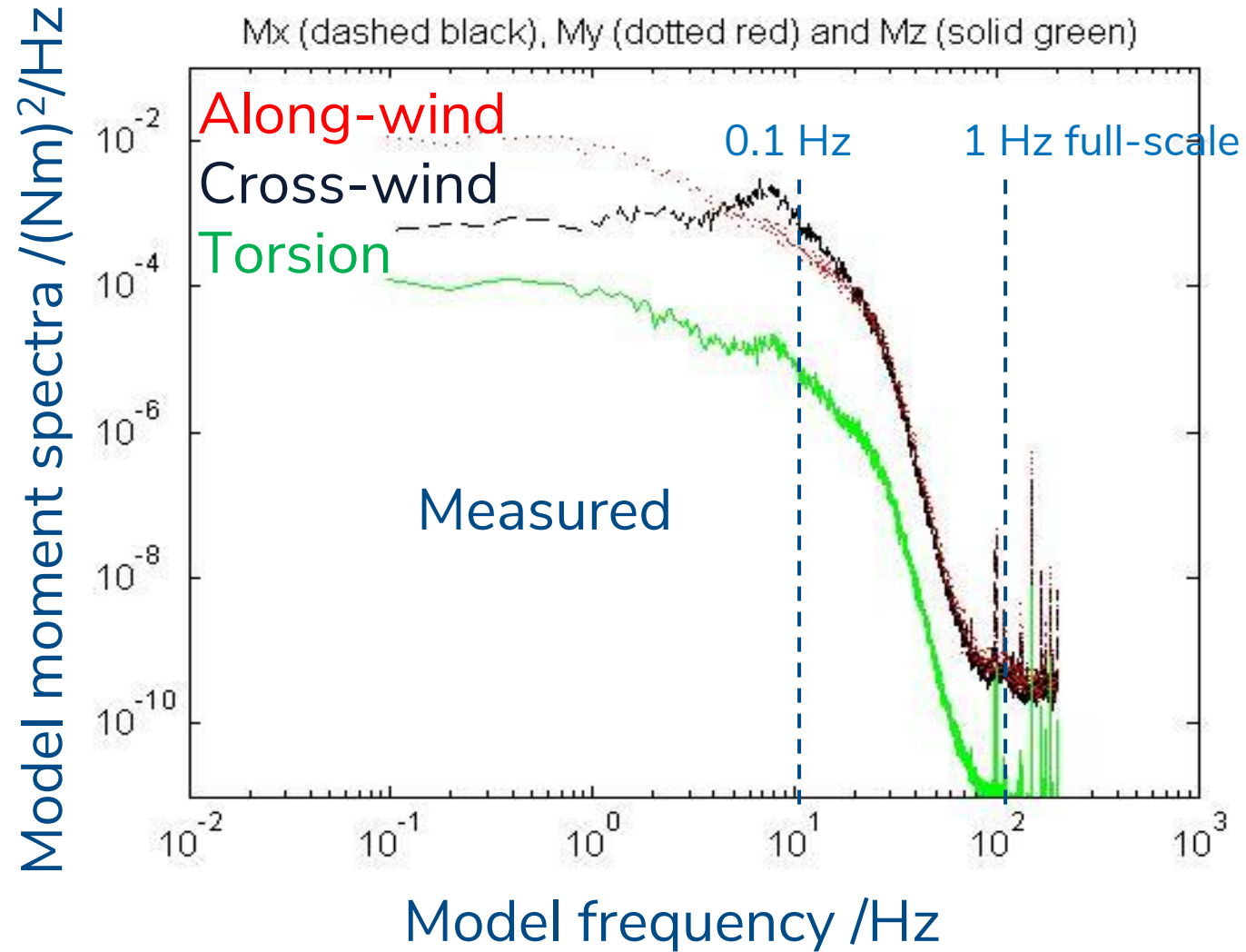


Integrated pressure

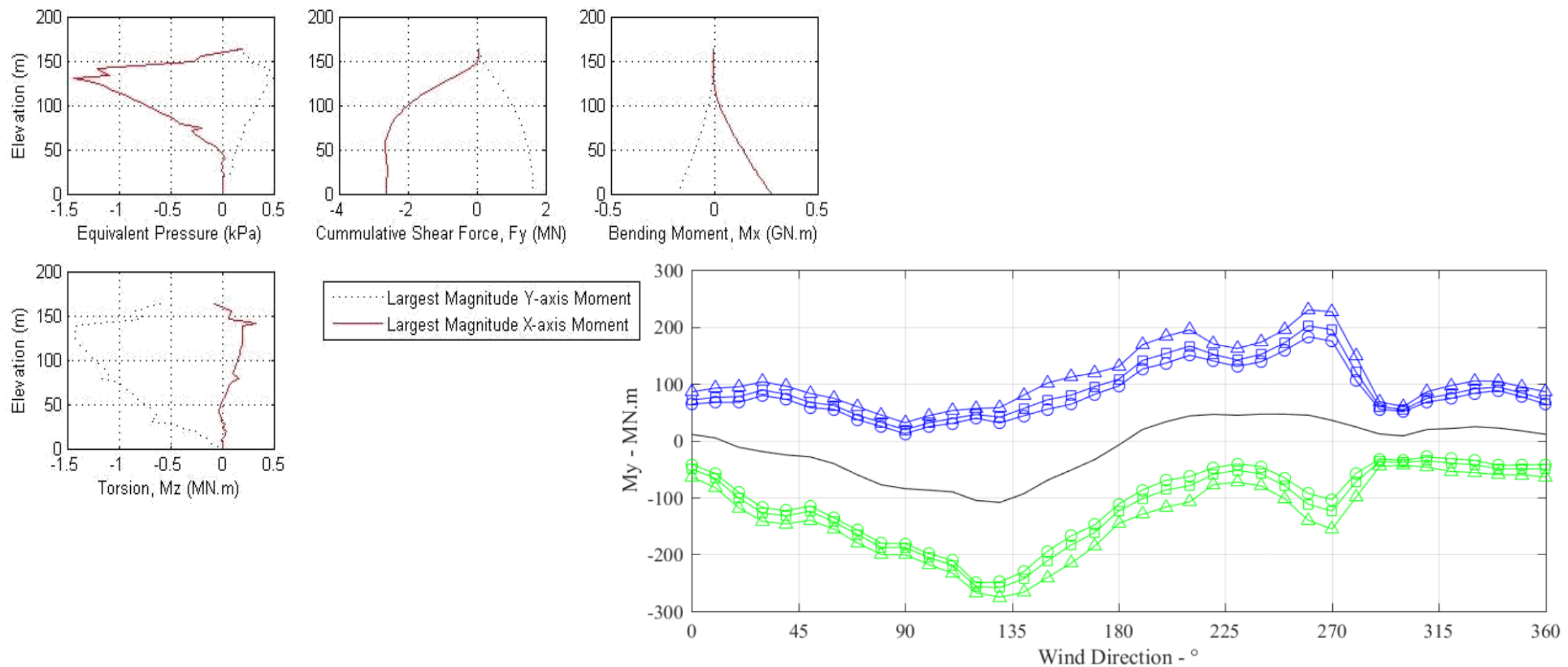
- Point pressures are integrated around the structure (need to be aware of area averaging difficult for complex geometries)
- Rigid model, no attempt to model aeroelastic properties of the structure
- Short testing time, but complex post processing
- Mode shape corrections are required
- Easy to recalculate response if structural dynamics properties change
- Gives more accurate force distribution up the height of the building



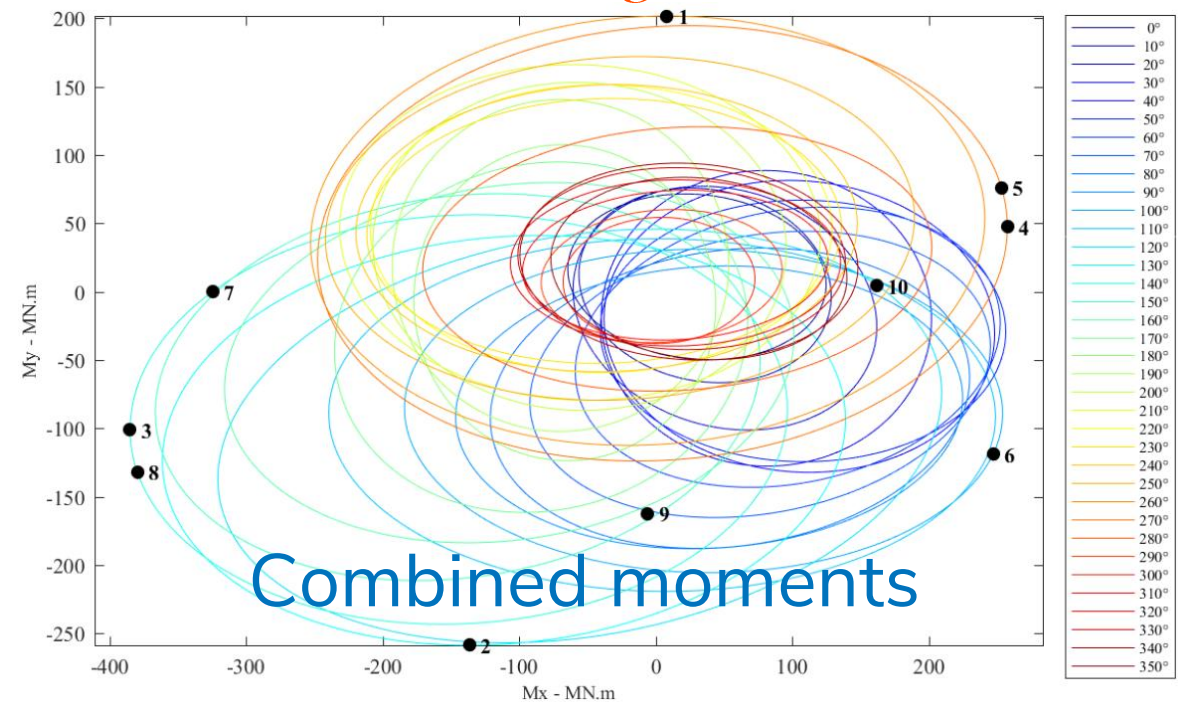
HFB / Integrated pressure



Typical Output



Structural loading

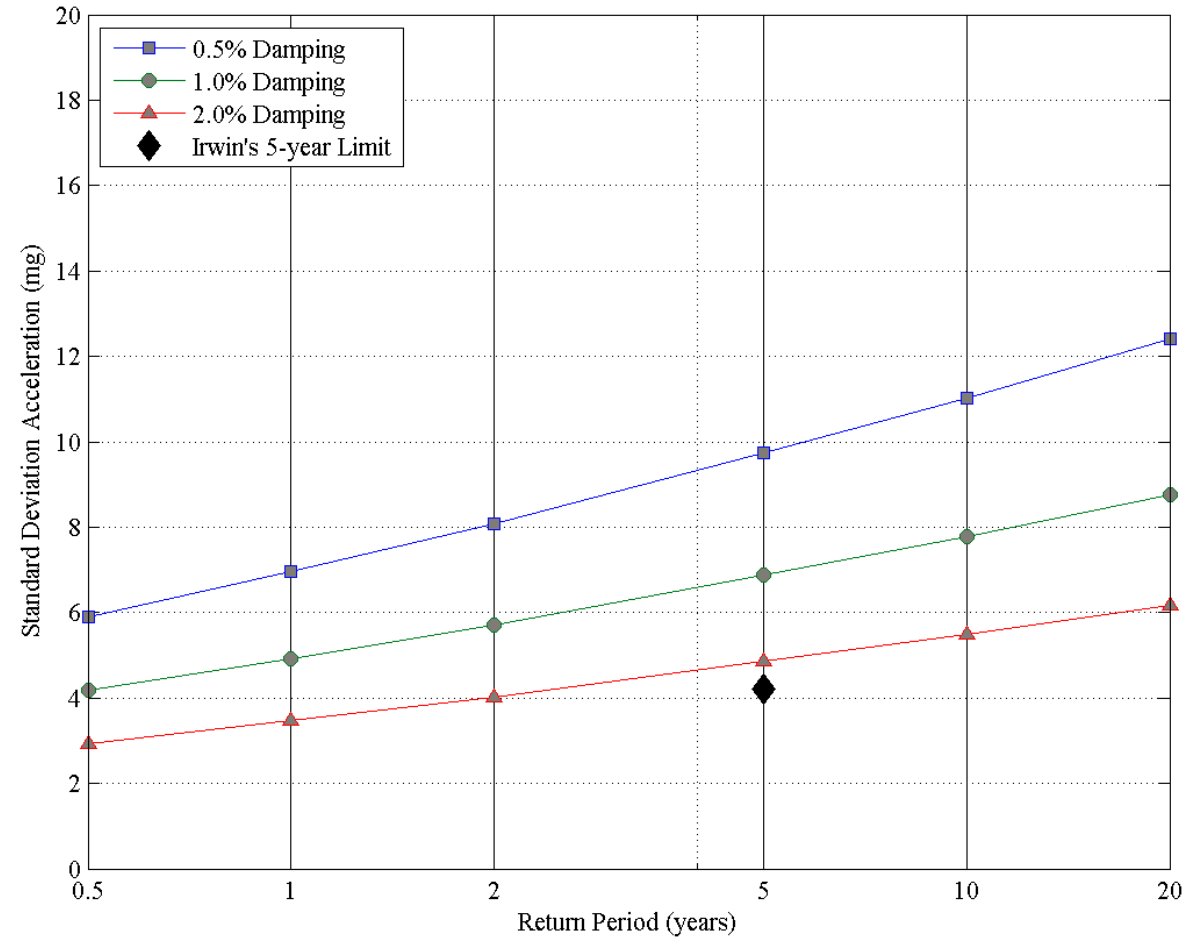
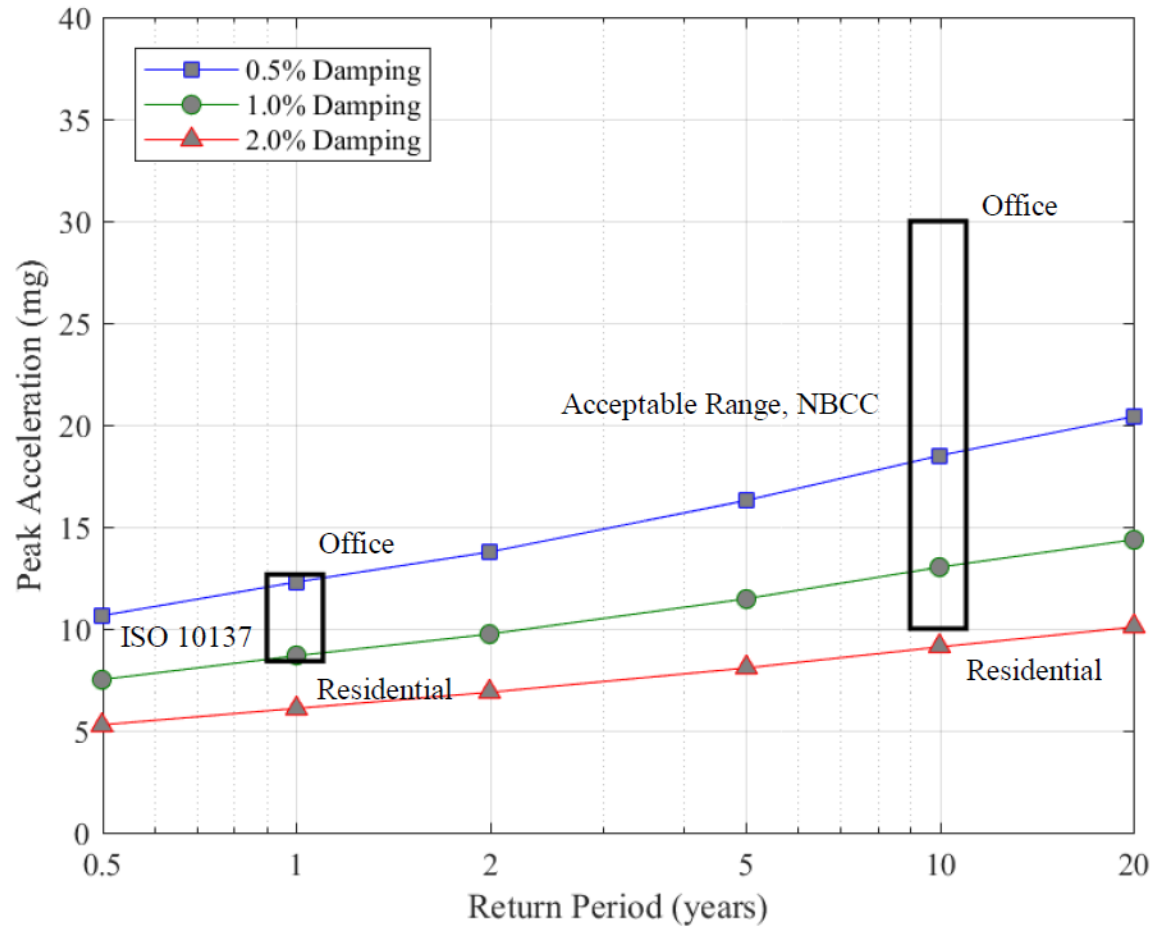


Load Definition		
Case	Comb.	Significance
1	A+	Max My, Vx
2	A-	Min My, Vx
3	B+	Min Mx, Max Vy
4	B-	Max Mx, Min Vy
5	C1	Max res. M w/ V in Q1
6	C2	Max res. M w/ V in Q2
7	C3	Max res. M w/ V in Q3
8	C4	Max res. M w/ V in Q4
9	D+	Max Mz
10	D-	Min Mz

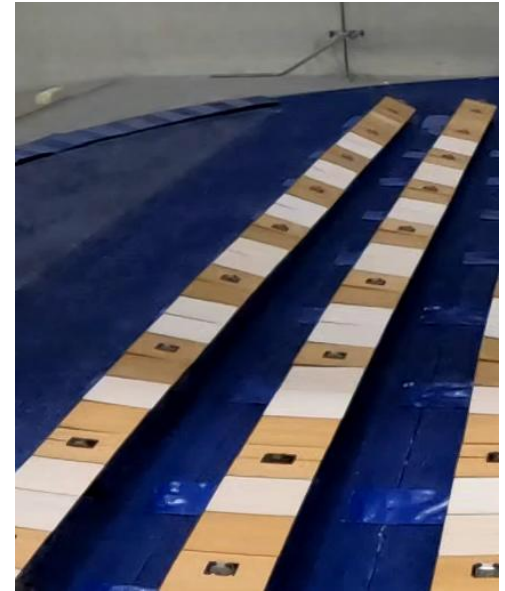
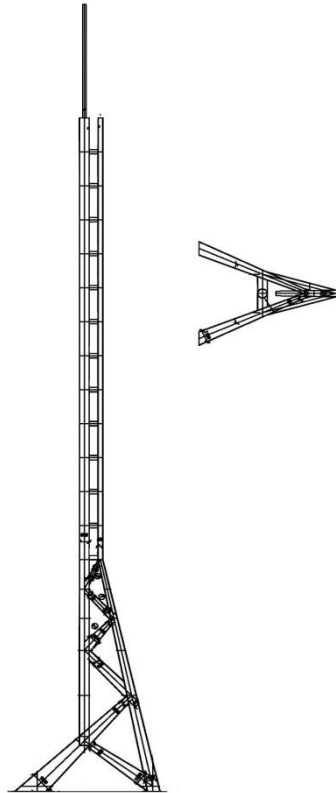
1000 Year Loads
Damping Ratio: 0.03
Natural frequencies, Hz
δx 0.380 δy 0.197 δz 0.778

Floor Label	Elev. (m)	Px (kN)			Py (kN)			Tz (MN-m)			Eccentricity (m)	
		Mean	SDEV	Peak	Mean	SDEV	Peak	Mean	SDEV	Peak	Ex	Ey
ROOF	154.57	40.3	24.2	131.3	5.4	37.5	-2.6	0.074	0.012	0.114	-0.02	-0.86
LEVEL_28	148.92	39.5	31.1	159.8	5.3	51.1	-5.7	0.072	0.011	0.111	-0.02	-0.69
LEVEL_27	145.27	25.0	32.3	151.9	3.3	53.4	-8.3	0.071	0.011	0.109	-0.04	-0.72
LEVEL_26	141.62	24.6	31.1	146.7	3.3	51.2	-7.8	0.070	0.011	0.107	-0.04	-0.73
LEVEL_25	137.97	24.3	29.7	140.9	3.2	48.7	-7.3	0.069	0.011	0.106	-0.04	-0.75
LEVEL_24	134.32	23.9	28.5	135.3	3.2	46.3	-6.9	0.068	0.011	0.104	-0.04	-0.77

Serviceability accelerations on top floor

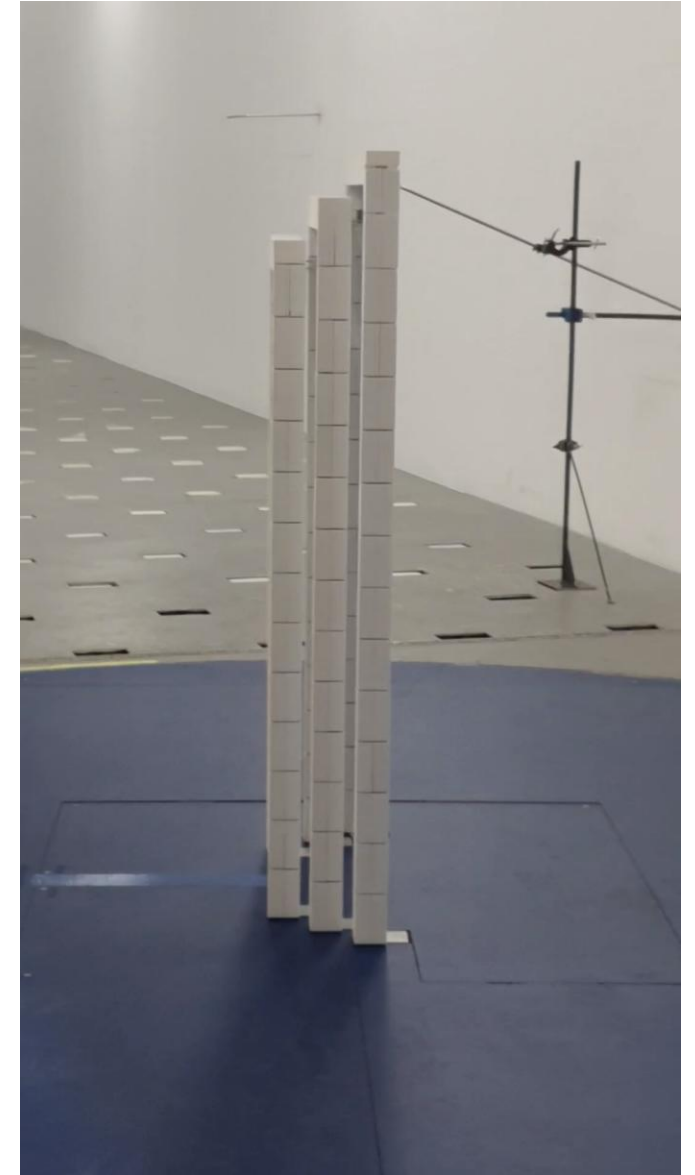


Full aeroelastic models (response)

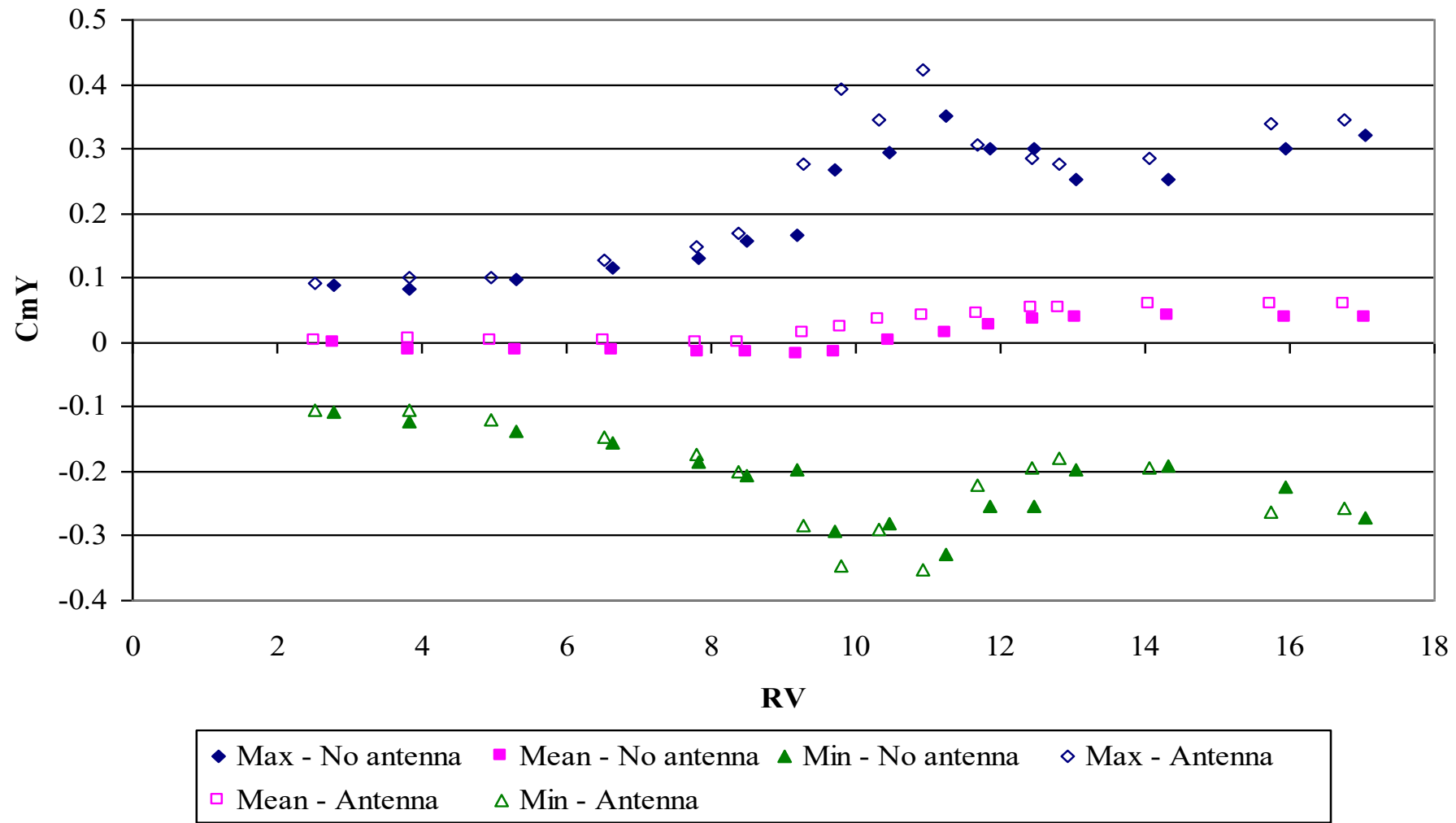


Full aeroelastic models

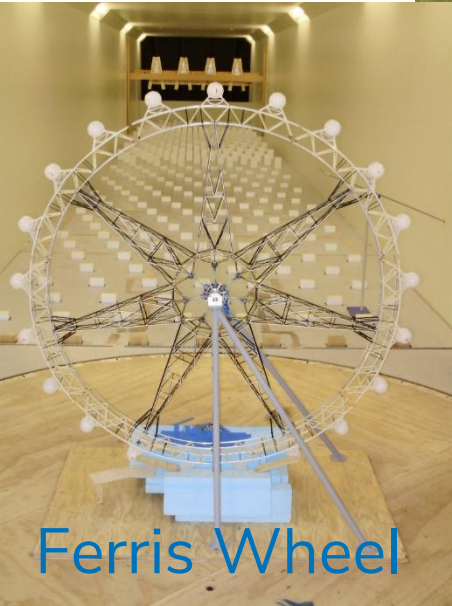
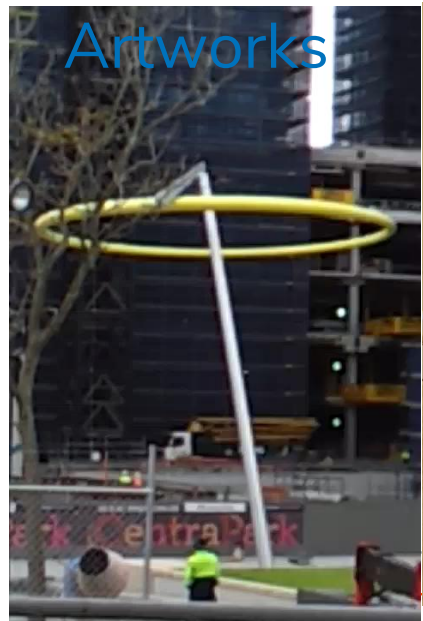
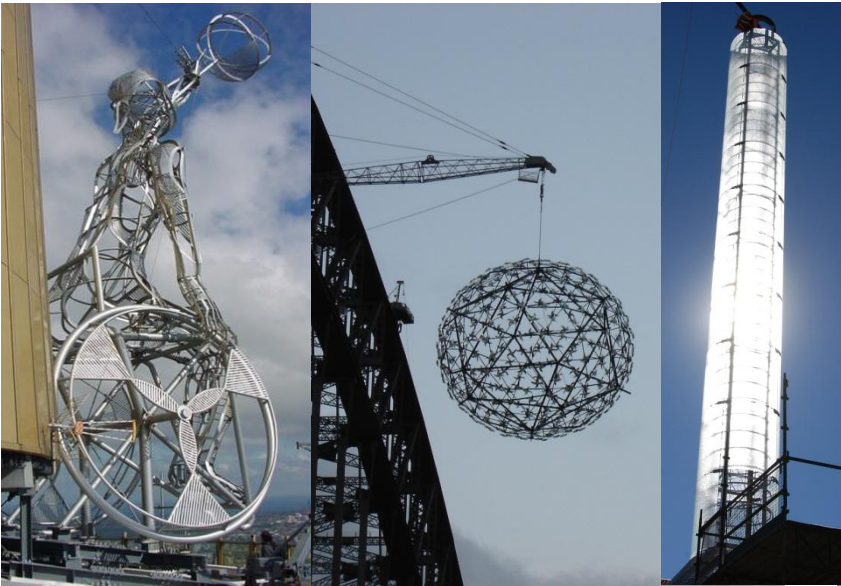
- Accurate mass and stiffness distribution
- Correct mode shapes
- Reproduce aeroelastic effects
- Complex, high precision unique models
- Difficult to incorporate damping
- Long test time, but simpler analysis
- Expensive



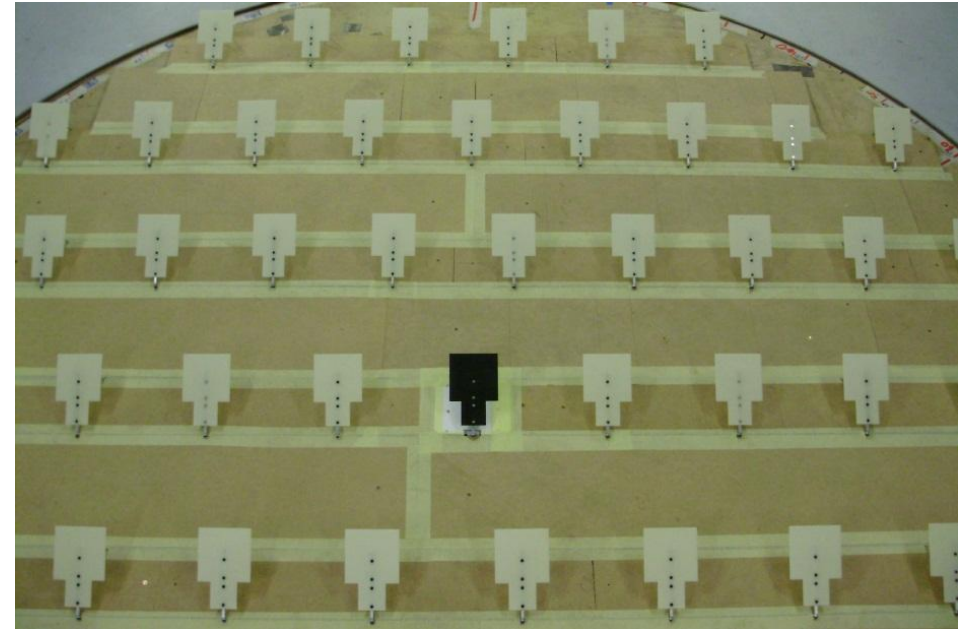
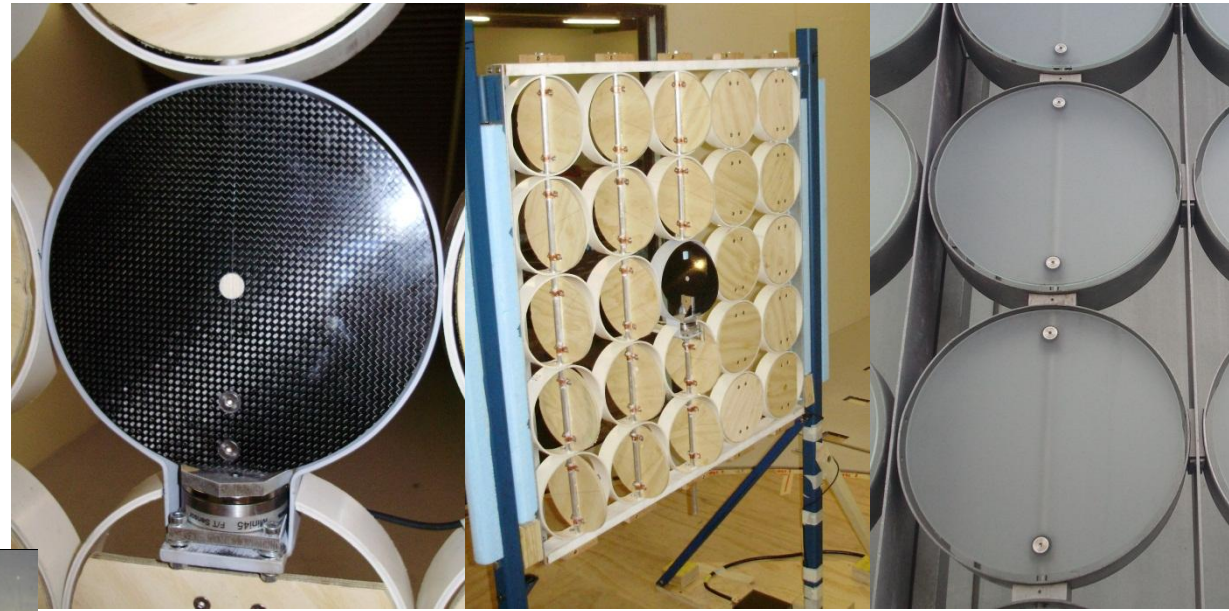
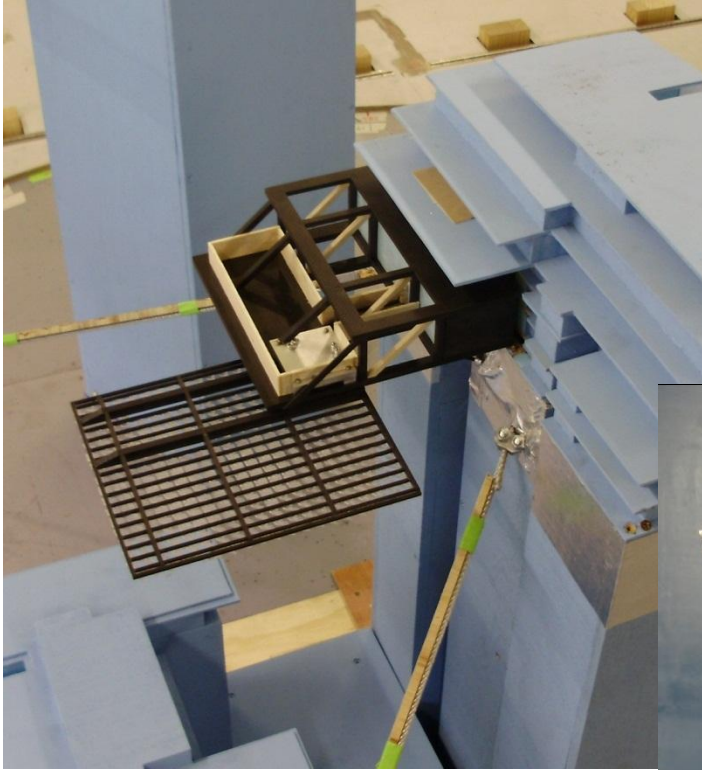
Typical results



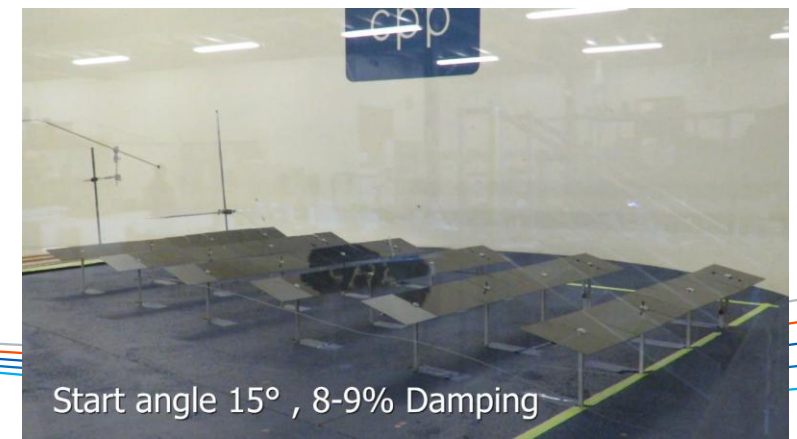
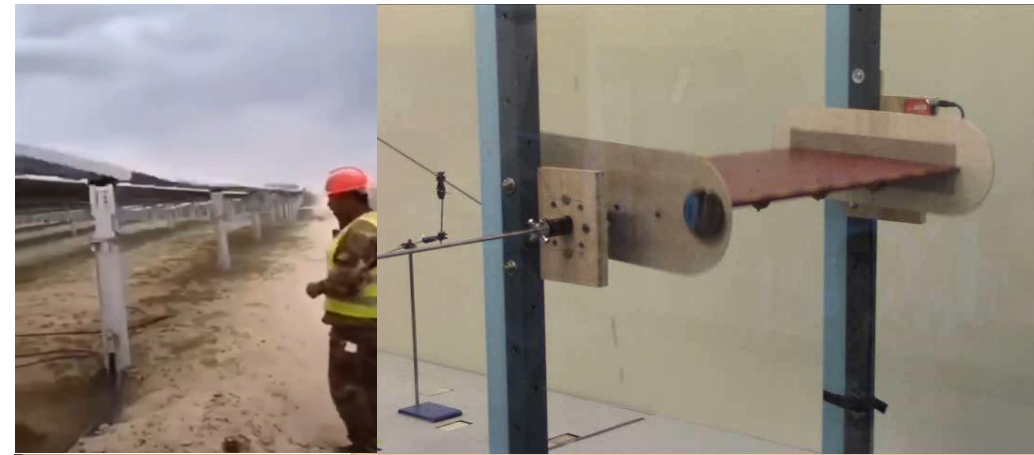
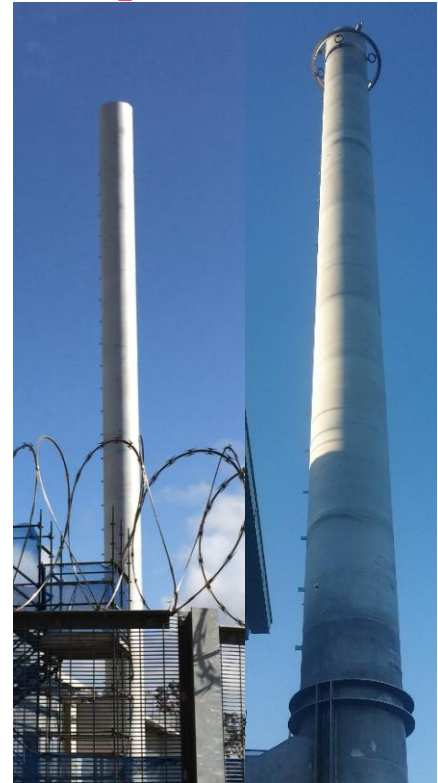
Other studies



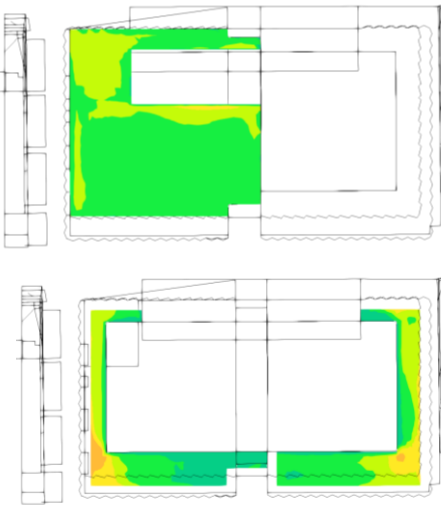
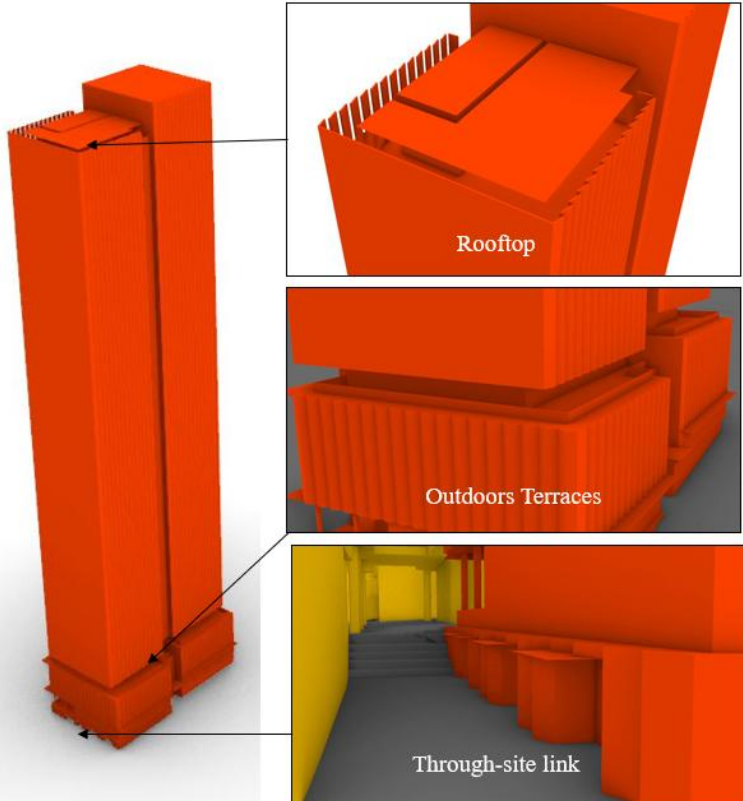
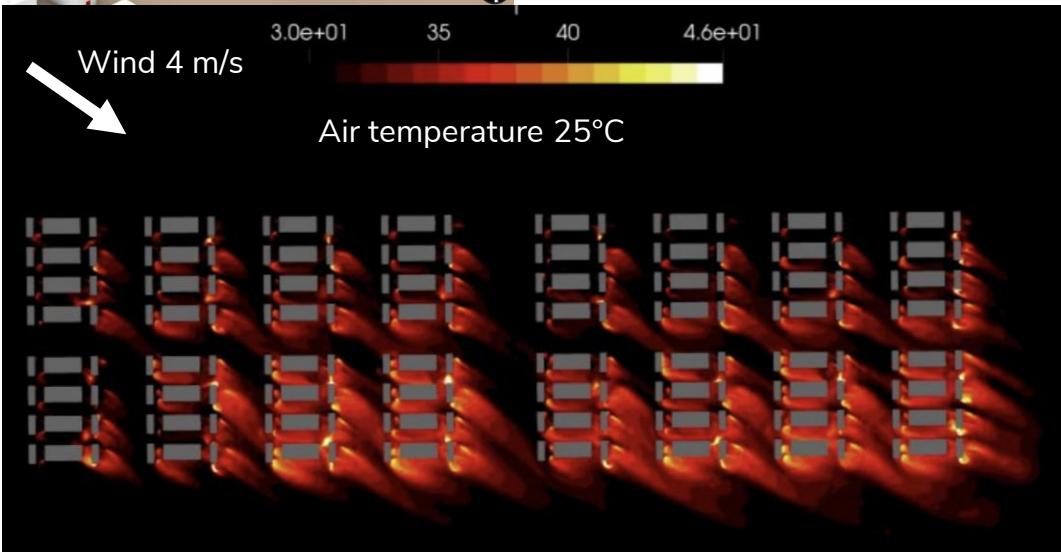
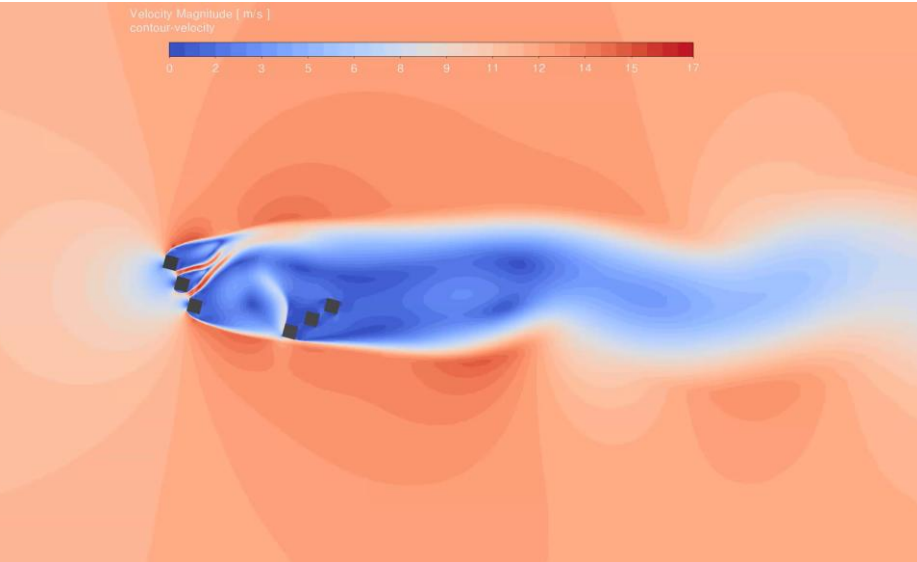
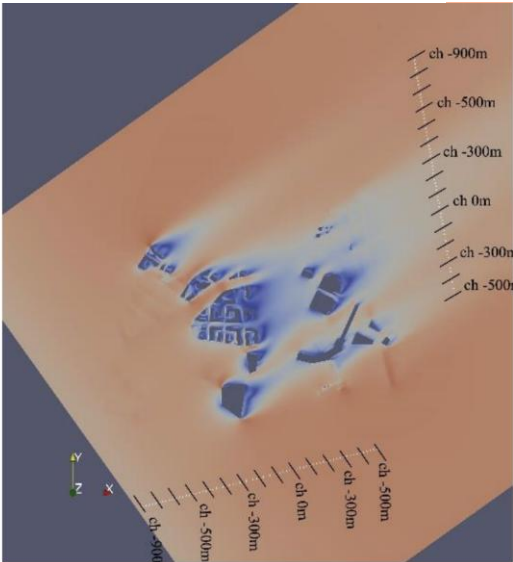
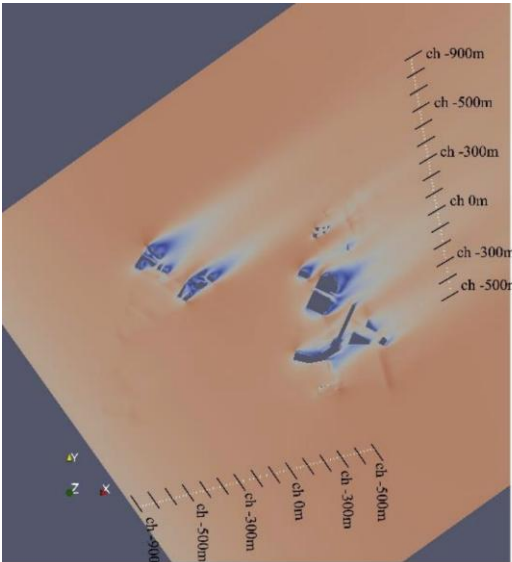
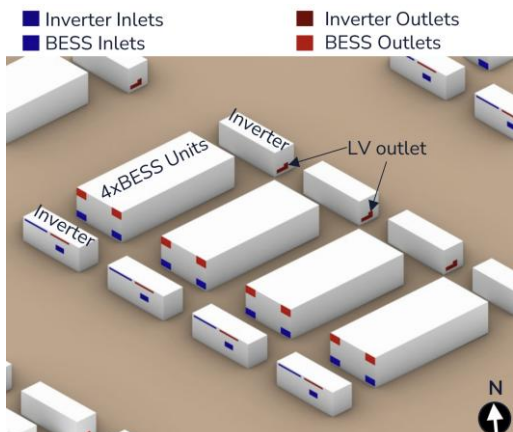
Quasi-steady and dynamic response



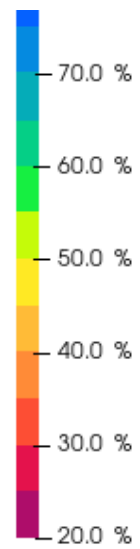
Dynamic response



CFD

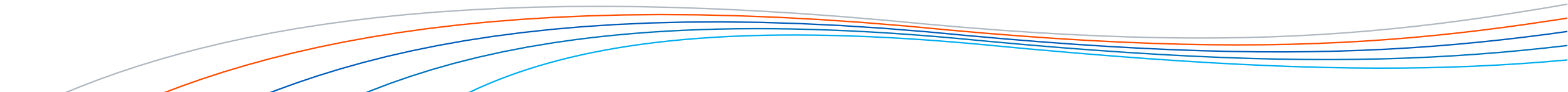


Annual (%)
for temp
(18-29.5°C)



Graduate Engineer

- Varied work
- Typical project less than 2 months
- Work with senior engineer to learn numerical/physical modelling techniques,
- Work in wind tunnel to learn instrumentation, experimental test techniques, and implications for analysis,
- Depending on interests move into detailed analysis, coding, business development, management...



Thanks for listening,
questions....

